

## BIOLOGY GRADE 10: PLANT TRANSPORT

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Subject             | Biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Strand              | Strand 2.0: Plant Anatomy and Physiology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Sub-Strand          | Sub-Strand 2.2: Plant Transport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Total Duration      | 8 lessons × 40 minutes = 320 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Overview of the plant transport system (xylem and phloem)</li> <li>– Root structure and function — root hair cells and absorption of water and mineral salts</li> <li>– Stem structure — external and internal features; arrangement of vascular bundles in herbaceous and woody stems</li> <li>– Leaf structure — internal anatomy; role in transport and gaseous exchange</li> <li>– Uptake and movement of water and mineral salts — osmosis, active transport, and the apoplast/symplast pathways</li> <li>– Transpiration — definition, factors affecting the rate, and its significance to water movement (transpiration pull)</li> <li>– Translocation — movement of manufactured food (sucrose and amino acids) through phloem; source-to-sink concept</li> <li>– Adaptations of plants to different water availabilities (xerophytes, hydrophytes, mesophytes)</li> </ul>                                                                                                                                                                                                                                                                 |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>describe the structure and function of the plant transport system, including the roles of xylem and phloem,</li> <li>explain how water and mineral salts are absorbed by roots and transported through the plant body,</li> <li>investigate and explain the factors that affect the rate of transpiration and its significance in water movement,</li> <li>explain the process of translocation of manufactured food substances from source to sink,</li> <li>relate the structural adaptations of plant tissues to their transport functions,</li> <li>appreciate the importance of plant transport processes to agriculture and food security in Kenya.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                             |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Critical Thinking and Problem Solving: The learner analyses why crops in Rift Valley farms wilt during dry spells and uses evidence from experiments (e.g., celery/white carnation dye experiments) to construct explanations about water movement in plants.</li> <li>– Communication and Collaboration: The learner works in groups to design and present experiments on transpiration using a potometer, shares findings with peers, and co-constructs annotated diagrams of vascular tissue.</li> <li>– Imagination and Creativity: The learner designs and draws cumulative models of the whole-plant transport system, revising them lesson by lesson as new evidence is gathered, and proposes creative solutions (e.g., irrigation scheduling) to reduce wilting in Kenyan farms.</li> <li>– Learning to Learn: The learner reflects on growing understanding by updating the Driving Question Board and self-assessing model revisions across the unit.</li> <li>– Self-efficacy: The learner develops confidence by independently designing a controlled transpiration investigation and presenting conclusions to the class.</li> </ul> |

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| <b>Core Values</b>                                | <ul style="list-style-type: none"> <li>– Responsibility: The learner handles laboratory specimens (real roots, stem cross-sections) and living plant material carefully and disposes of materials responsibly, reflecting stewardship of the natural environment.</li> <li>– Respect: The learner appreciates diverse contributions during group discussions on how Kenyan farmers manage water for crops, and respects peers' experimental observations even when results differ.</li> <li>– Unity: The learner collaborates across gender and ability groups when conducting transpiration experiments and constructing shared plant transport models.</li> <li>– Social Justice: The learner connects the science of plant transport to issues of food security and equitable access to water for smallholder farmers in regions such as Turkana and the semi-arid Eastern Province.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Asking Questions and Defining Problems: Students generate investigable questions from the anchoring phenomenon (wilting maize crops) and refine them into a class Driving Question Board throughout the unit.</li> <li>– Planning and Carrying Out Investigations: Students design controlled experiments to investigate factors affecting transpiration rate (light intensity, wind, humidity, leaf surface area) using improvised potometers built from locally available materials.</li> <li>– Analysing and Interpreting Data: Students collect quantitative data on water uptake rates, construct graphs, and identify patterns linking transpiration conditions to water loss in plants.</li> <li>– Developing and Using Models: Students iteratively build and revise a whole-plant transport diagram model — adding root absorption, xylem/phloem pathways, transpiration pull, and translocation — lesson by lesson.</li> <li>– Constructing Explanations: Students use osmosis, active transport, cohesion-tension, and pressure-flow concepts to explain how water and sugars move against or with concentration gradients.</li> <li>– Engaging in Argument from Evidence: Students use experimental data and microscopy observations to argue for or against proposed mechanisms of water movement (e.g., root pressure vs. transpiration pull).</li> <li>– Obtaining, Evaluating, and Communicating Information: Students research the impact of drought on Kenyan staple crops (maize, beans), evaluate sources, and communicate findings through annotated diagrams and oral presentations.</li> </ul> |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Environmental Conservation: The learner explores how deforestation around water catchment areas such as Mau Forest reduces transpiration rates and disrupts local water cycles, connecting plant transport science to conservation advocacy.</li> <li>– Agriculture and Food Security: The learner directly connects the mechanisms of water and nutrient transport in plants to the productivity of staple Kenyan crops (maize, beans, sukuma wiki) and to irrigation practices used by farmers in Mwea and Galana-Kulalu.</li> <li>– Climate Change: The learner investigates how rising temperatures and prolonged droughts (as experienced in northern Kenya) increase transpiration rates and accelerate wilting, linking plant biology to climate adaptation strategies.</li> <li>– Health Education: The learner recognises that disruption of plant nutrient transport (e.g., mineral salt deficiencies) reduces the nutritional quality of food crops such as ugali (maize flour) and sukuma wiki, with direct implications for community nutrition.</li> <li>– Safety and Security: The learner handles laboratory chemicals (iodine solution, eosin dye) and sharp tools (razor blades for stem sections) following proper safety protocols.</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Botany / Plant Science: Understanding plant vascular systems is foundational for careers in plant research at institutions such as KEFRI (Kenya Forestry Research Institute) or KALRO (Kenya Agricultural and Livestock Research Organisation).</li> <li>– Agronomy and Agricultural Extension: Knowledge of water uptake, transpiration, and nutrient transport informs irrigation scheduling, fertiliser application, and crop management advice given to Kenyan smallholder farmers.</li> <li>– Horticulture: Careers in Kenya's cut-flower industry (Naivasha) and fruit/vegetable export sector require understanding of post-harvest water relations and vascular tissue integrity to maintain produce quality.</li> <li>– Medicine / Pharmacognosy: Many Kenyan medicinal plants are used in traditional and modern medicine; understanding phloem transport of secondary metabolites (alkaloids, tannins) is relevant to pharmaceutical research.</li> <li>– Environmental Science and Conservation: Understanding transpiration and its role in the water cycle supports careers in watershed management, reforestation (e.g., Mau Forest restoration), and climate change mitigation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |

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|                                                 | <p>– Food Technology: Knowledge of how mineral salts and sugars are translocated in plants informs biofortification programmes aimed at improving the nutritional content of staple crops such as maize and beans.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Focus for Lessons</b>                        | <p>This unit focuses on how plants move water, mineral salts, and manufactured food substances through specialised vascular tissues (xylem and phloem), anchored in the real-world phenomenon of maize crops wilting on Kenyan farms during dry seasons. Students use a phenomenon-driven, inquiry-based approach — conducting hands-on experiments with local plant materials, building iterative models, and linking molecular-level processes (osmosis, active transport, cohesion-tension) to whole-plant and ecosystem-level outcomes. Throughout the unit, the science of plant transport is continuously connected to Kenyan agricultural contexts, food security, and environmental conservation, ensuring relevance and deep conceptual understanding.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Driving Question / Key Inquiry Questions</b> | <p>DRIVING QUESTION: How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. How are water and mineral salts transported in plants?</li> <li>2. How are manufactured food substances transported in plants?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Anchoring Phenomenon</b>                     | <p>ANCHORING PHENOMENON — "The Wilting Maize of Kibwezi":</p> <p>A farmer in Kibwezi, Makueni County — a semi-arid region of Kenya — notices that her maize crop wilts dramatically every afternoon during the dry season, even though she watered the soil that same morning. By evening, some plants recover; others do not. She also notices that the maize plants growing near a dry seasonal riverbed wilt far sooner than those on the lower, wetter slopes. Meanwhile, a neighbour's sukuma wiki plot, which is shaded by a large mango tree, barely wilts at all. The farmer is puzzled: the soil has water, she can see it — so why are the plants wilting? Why do some recover but others die? Why does shade seem to protect the sukuma wiki?</p> <p>This phenomenon is puzzling because it challenges students' intuitive belief that "watering = no wilting." It reveals that water availability in soil is not the same as water availability inside the plant, and that something must be actively moving water from soil through root, stem, and leaf — and losing it somewhere along the way. The scenario is directly familiar to Kenyan students, many of whom have observed wilting crops on family farms or school gardens, making it personally meaningful and emotionally engaging.</p>                                      |
| <b>Storyline Thread</b>                         | <p>Lesson 1 — Encounter the Phenomenon (The Wilting Maize of Kibwezi): Students view photographs and a short description of the wilting maize scenario in Kibwezi. They share personal observations from home farms and school gardens, record initial ideas about why plants wilt, and co-create the first Driving Question Board (DQB) with "What We Think We Know" and "What We Wonder" columns. Outcome: Students generate investigable questions and commit to initial predictions.</p> <p>Lesson 2 — What Do Plants Need to Survive? (Plant Needs and the Case for a Transport System): Students brainstorm what plants need (water, mineral salts, sunlight, carbon dioxide) and map where each resource enters the plant and where it is needed. They recognise that resources enter at one location but are used at another — establishing the logical necessity of a transport system. Students add new questions to the DQB. Outcome: Students articulate why an internal transport system is essential and identify it as the key to solving the wilting mystery.</p> <p>Lesson 3 — The Big Picture: Overview of the Plant Transport System (Xylem and Phloem): Teacher introduces xylem and phloem as the two transport highways of the plant using a whole-plant diagram. Students label diagrams and use coloured arrows to show</p> |

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|                             | <p>the direction of water/mineral salt movement (upward, xylem) vs. food movement (downward and upward, phloem). Students begin their cumulative model. Outcome: Students correctly identify the two vascular tissues and their transport directions.</p> <p>Lesson 4 — The Entry Point: Root Structure and Water Absorption: Students observe real roots of bean or maize seedlings (grown in the school garden) under a hand lens and, where possible, under a microscope. They draw and label root hair cells, the cortex, endodermis, and xylem. They link the large surface area of root hairs to efficient water and mineral salt absorption. Students conduct a simple osmosis demonstration (potato strips in salt solution vs. water) to confirm the mechanism. Outcome: Students explain how water enters the root by osmosis and how mineral salts enter by active transport, and add these mechanisms to their cumulative model.</p> <p>Lesson 5 — The Highway: Stem Structure and Vascular Bundles: Students cut cross-sections of herbaceous stems (e.g., sunflower or bean stems from the school garden) with a razor blade, stain with iodine or eosin, and observe vascular bundle arrangement. They compare herbaceous and woody stem diagrams. The coloured celery/carnation experiment is set up (results observed in Lesson 6). Students update their cumulative model to include stem pathways. Outcome: Students identify xylem and phloem within vascular bundles and describe their arrangement in herbaceous stems.</p> <p>Lesson 6 — The Destination: Leaf Structure and Transpiration: Students dissect a leaf (e.g., sukuma wiki or a Ficus leaf), observe the waxy cuticle, veins, and stomate positions (lower epidermis impression using clear nail varnish). They observe and record results of the coloured dye experiment from Lesson 5 — confirming xylem transport. The "sweating leaf bag" demonstration (tied around a branch in the school compound) is reviewed. Students define transpiration and list factors that affect its rate. Outcome: Students explain how water is lost through stomata and why the rate varies, and add leaf anatomy and stomata to their cumulative model.</p> <p>Lesson 7 — The Engine: Transpiration Pull, Cohesion-Tension, and Root Pressure: Students conduct a structured investigation using an improvised potometer (a cut stem, rubber tubing, capillary tube, and ruler) to measure water uptake under different conditions (fan blowing = wind, lamp = light, petroleum jelly on leaves = blocked stomata). They collect and graph data, then construct an explanation linking transpiration pull (cohesion-tension mechanism) and root pressure to upward water movement. Students return to the DQB and answer: "Why does the maize wilt even after watering?" — the rate of transpiration exceeds the rate of water absorption. Outcome: Students explain the cohesion-tension mechanism and use quantitative data to support it; they partially resolve the anchoring phenomenon.</p> <p>Lesson 8 — The Return Journey: Translocation and Final Model Build: Students examine the "girdled tree" photographs and the concept of source-to-sink translocation through phloem (sucrose, amino acids). They use the pressure-flow hypothesis to explain translocation direction. Students then complete their final cumulative model of the entire plant transport system — annotating xylem (water + minerals, root to leaf, transpiration pull), phloem (sugars + amino acids, source to sink, pressure flow), osmosis at root hairs, and stomatal transpiration. Groups present their models and return to the original Kibwezi wilting scenario to construct a full evidence-based explanation. DQB is reviewed: all driving questions are answered. Outcome: Students produce a complete, annotated whole-plant transport model and write a scientific explanation of the anchoring phenomenon, demonstrating mastery of all unit learning outcomes.</p> |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– The Coloured Carnation: A white carnation (or celery stalk) placed in red food-colouring solution turns red within hours — students observe coloured veins in the petals/leaves, raising the question of exactly which tissue carried the dye and how fast it travelled. (Supports: xylem structure and function, transpiration pull.)</li> <li>– The Sweating Leaf Bag: A clear plastic bag tied around a leafy branch of a tree in the school compound accumulates visible water droplets within 30–60 minutes on a sunny Nairobi morning, even though no water was added. (Supports: transpiration,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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|  | <p>stomatal function, environmental factors affecting water loss.)</p> <ul style="list-style-type: none"><li>– The Girdled Tree: Students examine photographs of trees in Karura Forest whose bark has been ring-stripped by poachers; the trees survive briefly but their leaves eventually yellow and die from the roots upward. (Supports: phloem translocation — cutting the phloem starves the roots of sugars.)</li><li>– The Drought-Resistant Cactus vs. the Wilting Bean: Side-by-side images of a cactus thriving in Turkana's arid landscape and a bean seedling wilting in a dry pot prompt students to ask: what structural features allow some plants to survive with almost no water? (Supports: plant adaptations — xerophytes vs. mesophytes.)</li></ul> |
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**ATTRIBUTION AND LICENSING**

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes): The Wilting Maize of Kibwezi: Encountering the Phenomenon

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                  |
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| <b>Purpose</b>              | To launch curiosity about plant water transport by confronting students with a puzzling real-world phenomenon from Kibwezi, Makueni County, that challenges the intuitive belief that watering a plant is sufficient to prevent wilting                                                                                                     |
| <b>Knowledge</b>            | Students will recall prior knowledge about plant needs and begin to recognise that water availability in soil does not guarantee water availability inside the plant                                                                                                                                                                        |
| <b>Skills</b>               | Students will observe photographic evidence critically, share personal farm and garden experiences, record initial ideas, generate investigable questions, and co-construct the first Driving Question Board                                                                                                                                |
| <b>Attitudes</b>            | Students will demonstrate curiosity, openness to uncertainty, and willingness to revise prior assumptions when confronted with puzzling evidence                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants?                                                                                                                                                                                                                                                                                      |
| <b>Purpose in Storyline</b> | This opening lesson anchors the entire unit. It introduces the central phenomenon — the wilting maize of Kibwezi — that students will return to in every subsequent lesson. By surfacing prior ideas and generating genuine questions, students develop ownership of the investigation and a personal reason to understand plant transport. |
| <b>Safety Notes</b>         | No laboratory hazards in this lesson. If printed photographs are distributed, ensure edges are not sharp. Students sharing personal farm experiences should be handled sensitively as some families may have experienced crop failure and associated economic hardship.                                                                     |

### B. LESSON OVERVIEW

Students are introduced to the anchoring phenomenon through photographs and a written scenario describing a Kibwezi farmer whose maize wilts in the afternoon despite morning watering. Students share personal observations from home farms and school gardens, record initial explanatory ideas individually, and then co-create the class Driving Question Board (DQB) with two columns: What We Think We Know and What We Wonder. The lesson ends with students committing to written initial predictions that will be revisited throughout the unit.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                       | Learner Experience                                                                                                                                                                                                                                         | Teacher Moves                                                                                                                                                                                                                                              | Sensemaking Strategy                                                                                                                                                                                                                                              | Formative Assessment Strategy                                                                                                                                                                                                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a><br><i>Related topic: not</i> | Students sit in groups of four. Before seeing any photographs, each student receives a prompt card that reads: A farmer in Kibwezi waters her maize early in the morning. By midday, the leaves are drooping. By evening, some plants stand upright again. | Distribute prompt cards and say: Read this story carefully. Think about any time you have seen a plant wilt — on your family shamba, at the school garden, anywhere. Write your own ideas first before talking to your group. [WAIT 3 MINUTES IN SILENCE — | Elicit-before-tell: activating prior knowledge before presenting the phenomenon ensures students connect new observations to existing mental models, making subsequent cognitive conflict more productive. Individual writing before group talk protects minority | Circulate and read student prompt cards during the silent writing phase. Look for: (1) students who write that wilting is caused by lack of water in soil — this is the target misconception to disrupt; (2) students who already mention heat or sun as a cause — affirm this as |

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| <p><i>an exact match for this lesson.</i><br/>Source: Seavuria<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms:<br/>plant transport<br/>wilting xylem</p>                                                                                                                                                      | <p>Others stay drooped and eventually die. The neighbouring sukuma wiki, growing in the shade of a large mango tree, looks healthy all day. Write down two things you think you already know about why plants wilt, and one question this story makes you want to ask. Students write independently for three minutes, then share within their group for two minutes. Groups do not report out yet — this preserves the diversity of ideas for the Observe phase.</p>                                                                                                                                                                                                                                                                                          | <p>circulate and read over shoulders to identify the most common prior ideas and the most surprising questions to highlight later.] Then say: Now share what you wrote with your group. Listen carefully — do others agree with you, or do they have different ideas? You do not need to reach agreement yet. [WAIT 2 MINUTES.] Do not take whole-class feedback yet. Say: Hold your ideas. We are about to look at some photographs that will give us more to think about.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>ideas from being suppressed by dominant voices.</p>                                                                                                                                                                                                                                                                                                                                        | <p>an important observation to investigate; (3) students who write questions about the inside of the plant — note these students to invite them to share during the DQB phase.</p>                                                                                                                                                                                                                                                                                                   |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: Seavuria<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms:<br/>plant transport<br/>wilting xylem</p> | <p>The teacher projects or pins up three photographs: (A) a close-up of wilted maize leaves curling inward in a dry Kibwezi field under bright midday sun; (B) a wider shot showing maize near a dry seasonal riverbed wilting more severely than maize on a lower, wetter slope; (C) lush, upright sukuma wiki growing under the canopy shade of a large mango tree, with visibly moist-looking soil around the maize in photograph A indicating the farmer did indeed water that morning. Students observe each photograph for 60 seconds silently, then annotate a printed observation sheet with three columns: What I See, What It Makes Me Think, What It Makes Me Wonder. After individual annotation, groups share observations for three minutes.</p> | <p>Display photograph A and say: Look carefully at this image. Do not explain it yet — just observe. What do you actually see? [WAIT 60 SECONDS IN SILENCE.] Then say: Now write in the first column — What I See. Only things you can actually see in the photograph, not your explanations yet. [WAIT 90 SECONDS.] Display photograph B and repeat. Then photograph C. After all three photographs, say: Now look at your What It Makes Me Think column. Here is a challenge — the soil in that photograph is damp. The farmer watered that morning. If the soil has water in it, why might the plant still be wilting? Does your original prediction still feel right? [WAIT 2 MINUTES for writing.] Then invite groups to share one surprising observation each. Use the phrase: What did you notice that surprised your group? Avoid evaluating responses — say: That is an interesting observation — let us keep it in mind as we build our questions.</p> | <p>Structured observation protocol (See-Think-Wonder) prevents students from jumping to premature explanations. The deliberate cognitive conflict — showing damp soil alongside wilted plants — is the key disruptive moment that makes the phenomenon puzzling and motivates the need for a transport system. Separating observation from explanation builds scientific practice habits.</p> | <p>Collect observation sheets at the end of the lesson or photograph them. Check whether students distinguish between observations and explanations — a key scientific skill. Listen during group sharing for the phrase the soil has water so why is it wilting — students who articulate this tension are ready to generate the most productive DQB questions. Students who still say it is wilting because there is no water need targeted questioning in subsequent lessons.</p> |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Students attempt to explain the phenomenon using only what they</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Say: You have observed the photographs and shared your</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Initial model drawing externalises mental models, making tacit ideas</p>                                                                                                                                                                                                                                                                                                                   | <p>Photograph all group diagrams before the gallery walk is</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| <p><a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Seavuria<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i></p> <p> <a href="#">Search ARES</a> for readings<br/> Search terms: plant transport wilting xylem</p>                                                                                                                                  | <p>currently know. Each group receives a large sheet of paper and draws a simple outline of a maize plant — roots, stem, leaves. They use arrows and labels to show what they think is happening to water on a hot afternoon in Kibwezi. Groups post their diagrams on the classroom wall. The class does a gallery walk (3 minutes) reading other groups diagrams silently before sitting down. Students then write an individual first draft explanation (3 sentences maximum) in their science notebooks answering the question: Why might a maize plant wilt even when the soil around it has water?</p>                                                                                                                                                                                                            | <p>ideas. Now I want each group to draw what you think is happening inside and outside the maize plant on a hot Kibwezi afternoon. Use arrows. Label what is moving and where it is going. You have four minutes. [WAIT 4 MINUTES — circulate and ask probing questions such as: Where does the water go after it enters the root? What happens to water in the leaves? Do not give answers.] After the gallery walk say: Look at the diagrams on the wall. Do they all agree? What is different between them? [WAIT 1 MINUTE.] Then say: Now write your own personal explanation — three sentences, in your notebook. Begin with the words: I think the maize wilts because... This is your prediction for the whole unit. We will come back to it at the end.</p>                                                                 | <p>visible to both teacher and student. The gallery walk creates low-stakes peer review and exposes students to alternative ideas without competitive evaluation. The individual written prediction creates a personal intellectual stake in the unit — students want to find out if they were right.</p>                                                                                                                | <p>complete. Look for: (1) diagrams that show water stopping at the root — indicates gap in understanding of whole-plant transport; (2) diagrams that show water leaving the leaf — indicates readiness for transpiration concepts in Lessons 6 and 7; (3) diagrams that include the sun or heat as a variable — affirm as scientifically important. Individual notebook predictions serve as a pre-assessment benchmark to compare with post-unit explanations.</p>                                                                             |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Seavuria<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i></p> <p> <a href="#">Search ARES</a> for readings<br/> Search terms:</p> | <p>The teacher reveals the unit Driving Question printed on a large card and posts it at the top of a dedicated DQB display: How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die? Each student receives three sticky notes. On the first sticky note they write one thing they think they know. On the second they write one question they genuinely wonder about. On the third they write what they think the most important thing to find out is. Students post their sticky notes in the correct columns of the DQB: What We Think We Know and What We Wonder. The teacher reads a selection aloud and clusters similar questions, inviting the class to suggest category</p> | <p>Post the driving question card and read it aloud. Say: This is the big question we will be investigating for the next several weeks. Every lesson will help us answer a part of it. Right now, we are going to fill our Driving Question Board — this is where we collect all our ideas and questions. It will grow every lesson. [Distribute sticky notes.] Say: On your first sticky note, write one thing you are fairly confident you already know about how plants get water. On the second, write a genuine question — something you really want to find out. On the third, write what you think is the single most important thing we need to understand to explain the Kibwezi wilting mystery. [WAIT 3 MINUTES IN SILENCE for writing.] Then invite students to come up and post notes in the correct columns. Read</p> | <p>The DQB is a living artefact of the class intellectual community. Clustering questions gives students agency in organising the inquiry. Naming clusters using student language rather than textbook language maintains authenticity. Returning to the DQB in every subsequent lesson creates coherence across the unit and signals that the teacher values student-generated questions as the engine of learning.</p> | <p>Read all sticky notes after class. Sort questions into three categories: (1) questions that will be directly answered in the unit — use these to inform lesson emphasis; (2) questions that reveal key misconceptions — e.g., Do plants drink like animals? or Does water go up the plant by itself? — address these explicitly in Lessons 4 and 7; (3) surprisingly sophisticated questions — e.g., How does water get to the very top of a tall tree? — celebrate these publicly in the next lesson to reward intellectual risk-taking.</p> |

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| plant transport<br>wilting xylem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | names for the clusters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | five or six aloud, clustering as you go. Say: I notice several of you are asking about what happens to water inside the stem — let us call that cluster The Inside Journey. Who else has a question that belongs there? Affirm all questions. Say: Every question on this board is a good scientific question. Our job this term is to answer all of them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Seavuria<br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES for readings</a><br>Search terms:<br>plant transport<br>wilting xylem | Students begin their cumulative unit model — a whole-plant diagram that they will add to in every lesson. Each student draws a simple outline of a maize plant in their science notebook (roots, stem with at least one vascular bundle indicated by a vertical line, and leaves with stomata indicated as dots on the lower surface). They label only what they currently know with confidence, and mark with a question mark any part they are unsure about. At the bottom of the model page, they write today the model shows what I already thought was true. By the end of this unit, I expect this model to show... and complete the sentence. The teacher explains that this model will be updated at the end of every lesson as their understanding grows. | Say: Scientists do not just ask questions — they also build models. A model is a drawing or diagram that shows how you think something works. We are going to build a model of the whole plant transport system across this entire unit. Today we start with what we know, or think we know. Draw a maize plant that fills a full page of your notebook — roots at the bottom, stem in the middle, leaves at the top. Label any part you are confident about. Put a question mark next to anything you are unsure of. [WAIT 4 MINUTES.] Then say: At the bottom of your model, write: Today my model shows... and describe in one sentence what your model currently represents. Then write: By the end of this unit, I expect my model to show... Give yourself a challenge. [WAIT 2 MINUTES.] Collect one example to photograph as a baseline. Say: We will return to this model at the end of every single lesson and add new understanding. By Lesson 8, your model should explain everything the Kibwezi farmer is experiencing. | Cumulative model building is the unit-long sense-making spine. Starting the model in Lesson 1 — even when students know very little — creates a concrete artefact that makes learning visible over time. Question marks on the model legitimise not knowing and reframe uncertainty as the beginning of investigation rather than a failure of knowledge. The student-authored prediction sentence creates intellectual investment in the completed model. | Photograph or collect baseline models before students leave. Do not grade them. Use them to: (1) identify students who already show water moving through roots, stem, and leaves — they may be ready for extension tasks in later lessons; (2) identify students who draw water only in the soil or only in the roots — these students need targeted scaffolding in Lesson 4; (3) note whether students place stomata — few will, but those who do show readiness for Lesson 6 content. Return models to students at the start of Lesson 2 for updating. |

## D. TEACHER REFLECTION

After this lesson, consider: Did all students genuinely engage with the puzzle, or did some dismiss it because they thought they already knew the answer? Were the photographs vivid enough to make the Kibwezi scenario feel real and personally relevant to students whose families farm? Did the DQB generate authentic questions from students, or were students writing what they thought the teacher wanted to see? Were there questions posted that surprised you — questions you had not anticipated? How will you use those questions to shape subsequent lessons? Did any student share a personal story about crop failure or wilting that could be honoured and woven into future lessons as a local data point? Reflect on wait time: did you give students full silence during the three-minute individual writing periods, or did you fill the silence with prompts? Extended silence is uncomfortable but essential for genuine thinking. Finally, note any student who expressed scepticism — saying for example that this is obvious or of course it wilts in the sun. These students need deeper cognitive challenge; the potometer investigation in Lesson 7 is likely to be the moment that most productively disrupts their certainty.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| What did I observe?                                          | We observed photographs of wilted maize crops in Kibwezi, Makueni County, where the soil was visibly damp from morning watering but the plants were still drooping by midday. We noticed that maize near the dry seasonal riverbed wilted sooner, while sukuma wiki growing in the shade of a mango tree stayed upright and healthy all day.                                                                    |
| What did I learn?                                            | We learned that our first instinct — that wilting means the soil has no water — is challenged by this scenario. The soil had water, yet the plants still wilted. This tells us that there must be something happening inside the plant that we do not yet fully understand. We also learned that shade, slope, and soil moisture all seem to affect how badly a plant wilts.                                    |
| How does this explain the phenomenon?                        | We cannot fully explain the wilting yet. We have recorded our first predictions and initial ideas, but we know these may be incomplete or incorrect. Our Driving Question Board shows the questions we need to investigate. By the end of this unit, we aim to construct a complete scientific explanation of what the Kibwezi farmer is observing — one that is supported by evidence from our investigations. |

## LESSON 2 (80 minutes): What Do Plants Need to Survive? — Mapping the Case for a Transport System

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To establish that plants require multiple resources that enter at different locations from where they are used, creating the logical necessity for an internal transport system — the key to explaining why the Kibwezi maize wilts.                                                                                                                                                                                                                                                                                                         |
| <b>Knowledge</b>            | Students identify the essential needs of a plant (water, mineral salts, carbon dioxide, sunlight/light energy) and map the specific location where each resource enters the plant and the location(s) where each resource is needed or used.                                                                                                                                                                                                                                                                                                 |
| <b>Skills</b>               | Students construct a resource-flow diagram (annotated plant outline) showing entry points and use points for each plant resource, and use this diagram to articulate — in their own words — why a transport system must exist inside the plant.                                                                                                                                                                                                                                                                                              |
| <b>Attitudes</b>            | Students appreciate that scientific reasoning involves identifying gaps between what we observe (water in soil) and what we need to explain (water in the leaf), and value the practice of following evidence to build logical arguments.                                                                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants? How are manufactured food substances transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | In Lesson 1, students observed the wilting maize phenomenon and generated initial questions. In Lesson 2, students zoom out to ask: what does a plant actually need, and where do those needs come from? By mapping resource entry points against resource use points, students discover the central tension — resources enter at one place but are consumed at another — and conclude that something inside the plant must be doing the moving. This conclusion becomes the engine that drives the rest of the evidence-gathering sequence. |
| <b>Safety Notes</b>         | No hazardous materials are used in this lesson. If students handle live plant specimens (maize seedling, sukuma wiki leaf), remind them to wash hands after handling soil or plant material. Ensure no student has a known plant allergy before passing specimens around.                                                                                                                                                                                                                                                                    |

### B. LESSON OVERVIEW

Students begin by revisiting the Kibwezi wilting maize scenario from Lesson 1 and the questions captured on the Driving Question Board (DQB). Through a structured brainstorm, they list everything a plant needs to survive, then use a large annotated diagram of a maize plant to physically map where each resource enters versus where it is used. The mismatch between entry point and use point creates a productive tension: if water enters at the root and is needed at the leaf, something must be carrying it. Students name this the transport system and connect it directly to the wilting mystery. New questions generated are added to the DQB. The lesson closes with students revising their initial mental model of the plant to include an internal pathway.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                         | Learner Experience                                                                                                                                                              | Teacher Moves                                                                                                                                                                                              | Sensemaking Strategy                                                                                                                                                                        | Formative Assessment Strategy                                                                                                                                                                         |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Photosynthesis</a><br>Source: Khan Academy (English - US curriculum) | Students are shown a large projected image of the Kibwezi farmer's wilting maize field alongside a healthy sukuma wiki plot shaded by a mango tree (same images from Lesson 1). | Display the two images silently for 30 seconds before speaking. Say: 'Before we talk, I want you to think like a farmer — like this woman in Kibwezi. What does her maize plant actually need to survive?' | Activating Prior Knowledge via Journal Prompt + Think-Pair-Share. This strategy surfaces students' existing mental models of plant needs before instruction, giving the teacher a formative | Teacher scans journal entries during pair-share. Observable indicator: student lists include at least 3 of the 4 key plant needs (water, mineral salts/nutrients, sunlight, carbon dioxide). If fewer |

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| <p><a href="#">Search ARES for similar videos</a></p> <p><b>READING:</b><br/>No closely matching reading in the ARES library for this activity.</p> <p><a href="#">Search ARES for readings</a><br/>Search terms: plant transport system needs</p>                                                                                                                                     | <p>Individually, students write a response in their science journals to the prompt: 'List everything you think a plant needs to stay alive and healthy. Next to each item, write WHERE the plant gets it from.' Students then share with a shoulder partner before a class discussion is opened.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Issue the journal prompt on the board. Allow 4 minutes of silent individual writing. After 4 minutes, say: 'Turn to your partner and share your list. You have 2 minutes — each person must speak.' After pair-share, cold-call 5 students (aim for gender balance and include at least one student who may have farm experience): 'Achieng, what is one thing on your list that your partner did NOT have?' Record all responses verbatim on the board under the heading: WHAT PLANTS NEED. Do not correct or affirm yet — simply record. Apply 10-second wait time after each response before moving to the next student.</p>                                                                                                                                                                                                                                                                                                                       | <p>baseline and giving students ownership of the starting point. Recording all answers without judgment models scientific openness and prevents early closure.</p>                                                                                                                                                                                                                        | <p>than half the class lists mineral salts/nutrients, flag this gap and address it explicitly in Phase 2. Listen for misconceptions such as 'plants get food from soil' — note these for targeted correction in the Explain phase.</p>                                                                                                                                                                                                                                       |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/><a href="#">Photosynthesis</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p><b>READING:</b><br/>No closely matching reading in the ARES library for this activity.</p> <p><a href="#">Search ARES for readings</a><br/>Search terms: plant transport system needs</p> | <p>Groups of four receive: (a) a large outline drawing of a maize plant (roots, stem, leaves, with a soil line drawn) printed on A3 paper, (b) a set of 12 colour-coded resource cards (three cards each for: WATER, MINERAL SALTS, CARBON DIOXIDE, SUNLIGHT/LIGHT ENERGY — each card states the resource name and one clue sentence), and (c) a resource reference sheet with four short factual paragraphs (one per resource) describing what it is, where the plant obtains it, and what the plant uses it for. Students read the reference sheet, then physically place each resource card on the diagram — green star stickers mark WHERE it enters, red arrow stickers mark WHERE it is used. Groups record observations in a two-column table: ENTRY POINT   USE POINT.</p> | <p>Distribute materials and say: 'You have 15 minutes. Your job is to be plant detectives. Read the reference sheet carefully — it has real science in it — then decide together where to place each card on your maize diagram. Use the green stickers for entry, red arrows for use. Every group member must be able to explain every card placement.' Circulate continuously. Stop at each group for 90 seconds. Ask probing questions: 'You placed water entering at the root — where does your diagram show it going next? Show me with your finger.' and 'Your group placed sunlight entering at the leaf — but where exactly on the leaf? Does it enter underground leaves too?' After 15 minutes, call time. Say: 'Hold up your diagrams.' Do a gallery scan — look for groups that have placed mineral salts entering at the leaf (a common misconception) and flag those groups for discussion. Select two contrasting group diagrams (one</p> | <p>Evidence Mapping on an Annotated Diagram. By physically placing resource cards on a plant outline, students make the abstract concept of resource location concrete and spatial. The two-column Entry Point vs Use Point table forces students to notice that entry and use are often NOT the same location — this productive tension is the core intellectual move of the lesson.</p> | <p>Teacher observes card placements during circulation. Observable indicator of understanding: water and mineral salts are placed entering at roots (below soil line); carbon dioxide is placed entering at leaves (stomata); sunlight/light energy is placed being absorbed at leaves. Observable indicator of misconception: mineral salts placed at leaves, or 'food' placed entering from the soil. Teacher notes which groups need targeted questioning in Phase 3.</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Photosynthesis</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport system needs</p> | <p>Whole class examines two projected group diagrams. Students identify where the entry point and use point are the same location and where they are different. The teacher introduces the key concept: when a resource enters at one location but is needed at another, something must transport it. Students complete a structured reasoning frame in their journals: 'Water enters the maize plant at the _____ (location). Water is needed at the _____ (location). Because these are different, the plant must have a _____ to move water from one place to the other. Without this system, the leaf would _____, which is exactly what we saw in _____.' Students then apply the same logic to mineral salts and to sugars manufactured in the leaf (teacher introduces: leaves make sugar using sunlight and CO<sub>2</sub> — where must that sugar travel to feed the root?). Class discusses: what happens to the Kibwezi maize if this transport system breaks down or is overwhelmed?</p> | <p>Project the two group diagrams side by side. Say: 'I want everyone to look at these two diagrams and identify one place where the entry point and the use point are NOT the same location. You have 10 seconds to find one.' [Wait 10 seconds in silence.] Cold-call 3 students: 'Kamau — tell me one mismatch you see.' After three responses, say: 'So we have a problem. Water is down here [point to root] but the leaf needs it up here [point to leaf]. Carbon is in the air up here, but the root needs energy down there. How does any of it travel?' [Wait 12 seconds.] Accept 2–3 responses, then say: 'Scientists call this the transport system. And this is exactly what the Kibwezi farmer's maize depends on every single afternoon.' Distribute the reasoning frame. Allow 6 minutes for silent individual completion. Cold-call 2 students to read their completed frames aloud. Correct gently: 'Almost — the leaf makes sugar, not food from the soil. Let's adjust that.' Pose the extension question: 'If the maize plant is losing water out of its leaves faster than the transport system can deliver it from the root, what happens?' Wait 15 seconds. Cold-call 1 student.</p> | <p>Structured Reasoning Frame (Claim–Evidence–Reasoning scaffold). The fill-in frame forces students to connect the spatial mismatch they observed on the diagram to the logical conclusion that a transport system must exist, and then explicitly link that conclusion back to the anchoring phenomenon. This is an NGSS Science and Engineering Practice: Constructing Explanations from Evidence.</p> | <p>Collect completed reasoning frames at the end of the phase (or photograph them). Observable indicator of proficiency: student correctly names roots as water entry point, leaves as water use point, names 'transport system' as the connector, and uses 'wilt' or equivalent as the consequence. Observable indicator of developing understanding: student identifies entry and use points correctly but cannot yet articulate the consequence for the Kibwezi maize. Use this data to group students for Lesson 3 peer support.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Photosynthesis</a><br/> Source: Khan Academy (English</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Each student receives two sticky notes. On the first (blue), they write one NEW question today's activity has raised about how the transport system works — something they do not yet know. On the second (orange), they write one thing they now understand</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Distribute sticky notes. Say: 'You have 3 minutes — silence please. Blue note: one question you now have that you did not have before. Orange note: one thing you can now explain about Kibwezi that you could not explain on day one.' After 3 minutes, direct students to</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Driving Question Board — Metacognitive Tracking. The two-colour sticky note system separates new uncertainty (blue) from consolidated understanding (orange), making students' growing knowledge visible to themselves and to the class.</p>                                                                                                                                                           | <p>Teacher reads all orange notes before the next lesson. Observable indicator: student can state at least one specific causal link (e.g., 'Water enters at the root and must travel to the leaf, so the plant needs a transport system'). If orange notes are vague ('I learned</p>                                                                                                                                                                                                                                                     |

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| <p>- US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> No closely<br/> matching reading<br/> in the ARES<br/> library for this<br/> activity.</p> <p> <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport<br/> system needs</p>                                                                                                                          | <p>about the Kibwezi wilting maize that they did NOT understand at the start of Lesson 1. Students post both notes on the class DQB under the appropriate columns: NEW QUESTIONS   GROWING UNDERSTANDING. The teacher reads selected notes aloud and clusters the new questions by theme (e.g., 'How exactly does water travel up?', 'What carries the sugar?', 'Why do some plants wilt more than others?'), previewing upcoming lessons.</p>                                                                                                                                                                                                                                                                             | <p>post notes on the DQB. Read 4 blue notes aloud — choose one about water movement, one about sugar, one about wilting, one unexpected. Say: 'These are the exact questions scientists asked when they first tried to understand plant transport. We are going to answer every single one of these over the next six lessons.' Point to specific new questions: 'This question — about how water moves up against gravity — that is Lesson 3. This one about what happens inside the stem — that is Lesson 4.' This explicit roadmapping builds anticipation and shows students their questions matter.</p>                                                                                                                                                                                                                                                                                                                 | <p>Clustering new questions by the teacher models scientific categorisation and previews the lesson sequence as a coherent investigation.</p>                                                                                                                                                                                                                                                                                                | <p>about plants'), teacher plans a brief whole-class re-voicing at the start of Lesson 3 to sharpen understanding before moving on.</p>                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Photosynthesis</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> No closely<br/> matching reading<br/> in the ARES<br/> library for this<br/> activity.</p> <p> <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport<br/> system needs</p> | <p>Students return to the annotated maize plant diagram from Phase 2 and revise it to become their first cumulative transport model. They add: (1) a bold upward arrow along the stem labelled WATER + MINERAL SALTS PATHWAY (use blue pencil/pen), (2) a bold downward arrow along the stem labelled SUGAR PATHWAY (use red pencil/pen), (3) a label at the leaf: WATER LOST HERE → WILTING IF TOO FAST, and (4) a legend. Under the diagram, students write one sentence: 'This model helps explain the Kibwezi wilting mystery because...' and complete it. Students do NOT yet label the specific tissues (xylem/phloem) — those come in Lesson 3. The model is kept in their science journal as a running record.</p> | <p>Say: 'Your maize diagram is about to become your first scientific model of plant transport. A model is not a drawing — it is an explanation. Every arrow, every label must mean something.' Demonstrate briefly on the projected blank diagram: draw one blue upward arrow and say, 'This arrow is not decoration. It represents real water molecules moving from this root cell, up through this stem, all the way to this leaf — right now, in every maize plant in Kibwezi.' Allow 8 minutes for model revision. Circulate and ask: 'If I removed your blue arrow from this diagram, what would happen to the maize in Kibwezi?' [Wait 10 seconds for student response.] Before closing, cold-call 2 students to read their sentence aloud. Affirm specificity: 'Excellent — you named the root and the leaf. That is precise scientific language.' Collect or photograph all models. Return at start of Lesson 3.</p> | <p>Cumulative Model Revision (NGSS Science and Engineering Practice: Developing and Using Models). By building the model in stages across lessons, students see science as a progressive, evidence-driven process rather than a fixed body of facts. The colour-coded pathway system (blue = water/minerals up; red = sugar down) builds the conceptual scaffold for xylem and phloem nomenclature in Lesson 3, reducing cognitive load.</p> | <p>Teacher reviews model diagrams and completion sentences before Lesson 3. Observable indicators of proficiency: diagram shows directionally correct pathways (water/minerals upward, sugar downward), water loss is labelled at the leaf, and the completion sentence explicitly connects the transport system to wilting. Observable gap: students who draw only one pathway (e.g., only water) need targeted questioning at the start of Lesson 3 about what the leaf produces and where it must send it.</p> |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully identify all four plant needs during Phase 1, or did mineral salts/nutrients remain invisible to most students? If mineral salts were absent from most lists, consider adding a short reading about Kenyan farmers using fertiliser (e.g., DAP fertiliser on maize in Rift Valley) at the start of Lesson 3 to cement this resource. (2) In Phase 2, how many groups correctly placed mineral salts entering at the root rather than the leaf? Groups that placed it at the leaf hold a persistent misconception (conflating 'nutrients sprayed on leaves' with normal mineral uptake) — plan a targeted 3-minute correction at the start of Lesson 3 using a soil-and-root photograph. (3) Were students able to articulate the 'entry point ≠ use point' tension independently, or did it only emerge after teacher prompting? If prompting was needed, the Phase 3 reasoning frame may need a partially-worked example added. (4) Review the DQB blue (new question) notes — do students' questions cluster around HOW water moves, or around WHY wilting happens? Questions about mechanism (how) signal readiness for Lesson 3 (xylem structure). Questions still anchored at the phenomenon level (why wilt) suggest students need one more phenomenon-grounding experience before moving to tissue-level explanation. Adjust Lesson 3 opening accordingly.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>What did I observe?</b>                   | Students observed (via annotated plant diagrams and resource cards) that different plant resources — water, mineral salts, carbon dioxide, and sunlight — enter the maize plant at different locations: water and mineral salts at the roots below the soil, carbon dioxide and sunlight at the leaves above the soil.                                                                                                                               |
| <b>What did I learn?</b>                     | Students learned that because plant resources enter at one location but are needed and used at a different location (e.g., water enters at the root but is needed at the leaf), a plant must have an internal transport system to move resources from entry point to use point — and that without this system, the leaf cannot get the water it needs and wilts, exactly as seen in the Kibwezi maize field.                                         |
| <b>How does this explain the phenomenon?</b> | Students can now begin to explain part of the Kibwezi wilting mystery: the maize plant wilts when water cannot reach the leaf fast enough through its transport system, even if water is present in the soil. The farmer's observation that 'the soil has water but the plant still wilts' makes sense because soil water and leaf water are separated by the entire length of the plant — and something must carry that water across that distance. |

## LESSON 3 (80 minutes): The Big Picture: Overview of the Plant Transport System (Xylem and Phloem)

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Purpose</b>              | To introduce students to the two vascular transport tissues — xylem and phloem — and establish the directional flow of water/mineral salts and manufactured food substances through the whole plant, directly connecting these transport highways to the wilting maize phenomenon in Kibwezi.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | By the end of this lesson, the learner should be able to: (a) identify xylem and phloem as the two vascular tissues responsible for transport in plants; (b) describe the direction of movement of water and mineral salts through xylem (soil → root → stem → leaf, upward); (c) describe the direction of movement of manufactured food substances through phloem (leaf → stem → root and other sinks, bidirectional but primarily downward); (d) locate xylem and phloem within a whole-plant diagram.                                                                                                                                                                                                                                                                           |
| <b>Skills</b>               | Labelling biological diagrams accurately; drawing and interpreting directional arrows to represent transport pathways; constructing an initial cumulative model of the plant transport system; comparing two transport systems using a graphic organiser; collaborative discussion and peer explanation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Attitudes</b>            | Appreciation of the elegant design of the plant vascular system; curiosity about how microscopic tissues explain observable phenomena such as wilting; responsibility in contributing to group diagram work; respect for peers' emerging ideas during model construction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Key Inquiry Question</b> | 1. How are water and mineral salts transported in plants? 2. How are manufactured food substances transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | In Lesson 1 students anchored on the mystery of the Kibwezi wilting maize and generated initial predictions. In Lesson 2 they mapped where each resource enters the plant versus where it is used, and concluded that an internal transport system must exist. Now, in Lesson 3, students zoom out to the whole-plant scale: they must understand WHAT the transport highways are (xylem and phloem), WHERE they run, and in WHICH DIRECTION each fluid moves — because without this 'road map', they cannot yet explain why water absorbed by roots in the morning fails to reach the leaves before afternoon wilting occurs. This lesson lays the structural scaffold upon which Lessons 4–8 will hang the mechanisms of transpiration pull, translocation, and stomatal control. |
| <b>Safety Notes</b>         | No hazardous materials are used in this lesson. Students handle printed diagrams and coloured pencils/markers only. Ensure classroom movement during group work is orderly to prevent trips or falls. If a digital projector is used, avoid directing the beam at students' eyes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### B. LESSON OVERVIEW

Students begin by briefly revisiting their Lesson 1 predictions and Lesson 2 resource-entry map, then pivot to the whole-plant scale. The teacher uses a large annotated diagram of a maize plant (rooted in Kibwezi soil) to introduce xylem and phloem as two distinct vascular 'highways'. Through a guided diagram-labelling activity, students use blue arrows for xylem flow and red arrows for phloem flow, building a directional map of the plant. A structured compare-and-contrast graphic organiser (Venn diagram) consolidates differences between xylem and phloem. Students then produce the first version of their cumulative whole-plant transport model, placing it in their science notebooks for revision in later lessons. The lesson closes by reconnecting to the Kibwezi farmer's puzzle: now that students know the two highways exist, the driving question sharpens — what force pushes water UP through xylem against gravity all the way to the topmost maize leaf?

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                            | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br>Source: Seavuria<br> Search ARES<br>for similar videos<br><br> <b>HTML:</b><br><a href="#">Vascular Tissue of Plants - Advanced (Flexbook)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> Search ARES<br>for similar readings | Students individually read a short two-sentence scenario card placed on their desks: 'The Kibwezi farmer cut a cross-section of a wilting maize stem. She saw two different kinds of tubes inside. One set of tubes was carrying water upward; another set seemed to carry something else in a different direction.' Students write a 2-minute silent prediction in their notebooks answering: (a) What do you think the two types of tubes are called? (b) Which direction does each fluid travel? (c) How might a blockage in one of these tubes explain the maize wilting even after morning watering? Students then share predictions with a shoulder partner (1 minute). Teacher cold-calls 4 students — two from different ability groups — to share predictions aloud. Teacher records key vocabulary that students already possess (e.g., 'veins', 'pipes', 'channels') on the Driving Question Board without correcting misconceptions yet. | Distribute scenario cards as students settle. Say: 'Read the card silently — you have 90 seconds. Then write your three predictions. Do NOT look anything up yet; I want YOUR thinking.' [Wait 2 minutes.] 'Now share with your shoulder partner — one person speaks, one listens, then switch. You have 60 seconds each.' [Wait 2 minutes.] Cold-call: 'Akinyi, what did you predict the two tubes are called?' [Wait 10 seconds.] 'Omondi, do you agree with Akinyi, or did you predict something different?' [Wait 10 seconds.] 'Wanjiku, what direction did you think the water moves — and why?' [Wait 12 seconds.] 'Kamau, how did you connect a blocked tube to the wilting maize?' [Wait 15 seconds.] Record student language on the DQB under 'What we THINK we know'. Say: 'Hold onto these predictions — by the end of today you will be able to check them against real biology.' | Strategy: Prediction-with-Justification. Students are required not just to guess but to give a reason, activating prior schema from Lesson 2 (resource entry and use points, water movement) and everyday observations of plant 'veins'. This surfaces misconceptions (e.g., 'all tubes carry the same thing upward') that the lesson will directly address, and creates cognitive dissonance that motivates learning.                                          | Observable indicator: Student notebooks show three written predictions with at least one justification linked to the wilting phenomenon. Teacher circulates during partner share and notes (on a clipboard checklist) which students correctly anticipate directional flow vs. those who believe water moves in all directions through one system — this informs cold-call selection and targeted questioning during Phase 3.                                                                                                      |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br>Source: Seavuria<br> Search ARES<br>for similar videos<br><br> <b>HTML:</b><br><a href="#">Vascular Tissue of Plants - Advanced (Flexbook)</a><br><i>Related topic: not</i>                                                                                                                                                                     | Activity 2A — Whole-Plant Diagram Annotation (25 min): Each student receives a large printed black-and-white diagram of a whole maize plant (roots in labelled Kibwezi soil, stem, leaves, developing maize cob). The diagram includes callout boxes for: root hair zone, root cortex, stem cross-section, leaf cross-section, and leaf surface with stomata. Using the teacher's projected annotated master diagram and a two-page reading excerpt (provided on the ARES platform, adapted from the KICD Grade 10 Biology textbook), students: (1)                                                                                                                                                                                                                                                                                                                                                                                                  | Distribute diagram sheets and reading excerpts simultaneously. Project the master annotated diagram on the board — but cover the labels with sticky notes initially. Say: 'Before I reveal the labels, use the reading to find the answers yourself. You have 10 minutes to label your diagram and draw your arrows. I will circulate — I am looking for correct arrow DIRECTION, not perfect art.' [Circulate for 8 minutes, noting errors without correcting immediately. Make specific targeted comments: 'Fatuma, your blue arrow is going the right                                                                                                                                                                                                                                                                                                                                      | Strategy: Annotated Diagram with Colour-Coded Evidence Trails. Using two distinct colours (blue/red) forces students to mentally separate two transport systems that share the same stem — a common point of confusion. Annotation in students' own words (not copied text) requires processing, not transcription. The progressive reveal of the master diagram provides immediate feedback without removing the productive struggle of independent labelling. | Observable indicators: (1) Diagram shows correctly placed labels for xylem and phloem in ALL four callout boxes (root, stem, leaf cross-section, leaf surface). (2) Blue arrows are strictly unidirectional upward; red arrows are bidirectional with one arrow toward root AND one toward the maize cob. (3) Reading highlights are colour-coded correctly. Teacher collects 6 diagrams (two high, two mid, two emerging students identified from Lesson 2 data) at the end of this phase for rapid review before Phase 3 begins. |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>label xylem and phloem in each callout box; (2) draw BLUE arrows showing the direction water and mineral salts travel through xylem (upward from root tip to leaf); (3) draw RED arrows showing the direction manufactured food (sugars) travel through phloem (from leaf downward to roots and also to the developing maize cob as a sink); (4) annotate in their own words what each tissue carries. Activity 2B — Structured Reading &amp; Highlight (10 min): Students read the excerpt independently, highlighting in BLUE every sentence that describes xylem structure or function, and in YELLOW every sentence about phloem. They write one 'surprising fact' in the margin.</p> | <p>direction — now check where xylem sits in the stem cross-section callout box.' 'Baraka, you've drawn the red arrow going only downward — re-read paragraph three carefully.']. After 10 minutes: 'Let's check together.' Reveal one callout box at a time on the projected diagram. Say: 'Look at the root hair callout — who has the label in the same position as mine? Raise your hand.' [Count hands.] 'If you had it differently, correct it now in a different colour so you can see your own thinking shift.' Repeat for each callout box. For phloem direction: 'I want to challenge you — raise your hand if you drew the red arrow going BOTH toward the root AND toward the maize cob.' [Wait 10 seconds.] 'That is correct — this is called bidirectional movement. We will explain WHY next lesson, but for now mark it on your diagram.' After Activity 2B: Cold-call 3 students: 'Njeri, read me your surprising fact.' [Wait 10 seconds each.]</p> |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> Source: Seavuria<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Vascular Tissue of Plants - Advanced (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i></p> | <p>Activity 3A — Venn Diagram Compare &amp; Contrast (12 min): In pairs, students complete a printed Venn diagram comparing xylem and phloem across six criteria: (i) type of cells (living/non-living); (ii) what is transported; (iii) direction of transport; (iv) energy requirement (passive/active); (v) location in the stem (relative position); (vi) connection to the Kibwezi wilting problem. Students use their annotated diagrams and the reading excerpt as reference. Activity 3B — Jigsaw Share-Out (8 min): Each pair joins another pair to form a group of four. Each pair takes one system (xylem OR</p>                                                                  | <p>For Activity 3A: 'You and your partner have 12 minutes to fill the Venn diagram. The hardest row is row four — energy requirement. The reading excerpt gives you a clue in paragraph five. Find it.' [Circulate. Targeted prompt if stuck: 'Think about what you know about osmosis — does osmosis need energy?'] For Activity 3B: 'When you teach your system, you must tell your partners: WHAT it carries, WHICH direction, and HOW that relates to the wilting maize. You have 4 minutes each.' [Monitor groups. If a group finishes early: 'Add a seventh row to your Venn diagram — what would</p>                                                                                                                                                                                                                                                                                                                                                           | <p>Strategy: Jigsaw with Phenomenon Reconnect. The jigsaw structure ensures every student must deeply understand one system well enough to teach it — teaching is the highest level of processing. The phenomenon reconnect at the close prevents the lesson from becoming abstract anatomy; it forces students to apply the structural knowledge to the observable puzzle of wilting, maintaining the storyline thread.</p> | <p>Observable indicators: (1) Venn diagram contains accurate entries for at least 5 of 6 criteria — particularly correct on 'direction of transport' and 'living/non-living cells'. (2) During jigsaw, teacher listens for correct use of terms 'xylem', 'phloem', 'upward', 'bidirectional', and 'turgor' (introduced informally). (3) During reconnect discussion, at least 70% of students can verbally link xylem blockage to wilting with a correct causal chain. Teacher records misconceptions heard during jigsaw (e.g., 'phloem only goes down') on a sticky note for targeted correction at the start of Lesson 4.</p> |

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| <p>Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>phloem) and 'teaches' it to the other pair using only their Venn diagram — no reading allowed. The listening pair asks at least one question. Activity 3C — Phenomenon Reconnect Discussion (5 min): Teacher leads a whole-class discussion tying today's evidence to the Kibwezi farmer. Teacher poses: 'Now that you know the two transport highways — if the xylem of a maize plant becomes blocked or cannot pull water fast enough, what would happen to the leaf?' Students respond and teacher links their answers to the afternoon wilting pattern.</p>                                                                                                                                                                                                                                                                                                                                                                                                           | <p>happen to the plant if ONLY this tissue was destroyed?'] For Activity 3C, pose the reconnect question. [Wait 15 seconds of silence before cold-calling — enforce the wait.] Cold-call: 'Ochieng, what would happen if xylem was blocked?' [Wait 10 seconds.] 'Adhiambo, do you agree? What would you add?' [Wait 10 seconds.] Summarise on the board: 'Blocked xylem = water cannot reach leaves = turgor pressure drops = wilting. This is EXACTLY what the Kibwezi farmer is seeing — but WHY does this happen on a hot afternoon? That is our next question.'</p>                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a><br/> Source: Seavuria<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Vascular Tissue of Plants - Advanced (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each student writes TWO sticky notes: (1) a BLUE sticky note stating one thing they now know about the plant transport system that they did NOT know at the start of the lesson (evidence-based claim, e.g., 'Xylem carries water upward through dead cells — no energy needed by the plant for this'); (2) an ORANGE sticky note stating one NEW question that today's lesson has raised for them (e.g., 'If xylem is made of dead cells, what force pulls the water all the way up to the top leaf of a tall maize plant?'). Students post their sticky notes on the class DQB under two columns: 'Evidence we now have' (blue) and 'New questions we need to answer' (orange). Teacher reads 3–4 orange notes aloud and says: 'These are exactly the questions that real plant biologists ask. Lessons 4 and 5 will give us the evidence to answer them.' Teacher draws a visible arrow on the DQB connecting today's blue notes to the original driving question,</p> | <p>Distribute sticky notes (or direct students to tear notebook paper into two pieces and colour the edges). Say: 'Blue note first — write one EVIDENCE-BASED claim. It must include the word xylem OR phloem. You have 90 seconds.' [Wait 90 seconds.] 'Orange note — write the most burning question today's lesson has left you with. If you have no questions, you are not paying close enough attention.' [Wait 90 seconds.] 'Come up row by row and place your notes.' After posting: Read 3 orange notes aloud expressively. Say: 'Whoever wrote this — thank you. This question is going to drive our next two lessons. Leave it on the board where everyone can see it.' Point to the DQB driving question banner: 'How does water and food travel through an entire plant and why does this journey sometimes fail?' Say: 'We now know WHAT the highways are. Our next question is: what is the ENGINE that drives water up the xylem</p> | <p>Strategy: Claim-Evidence-Question DQB Protocol. Requiring students to write an evidence-based claim (not just 'I learned about xylem') builds the habit of grounding knowledge in evidence — a core science practice. The new-question sticky note externalises metacognition and creates a living, visible record of the class's growing understanding across the 8-lesson sequence.</p> | <p>Observable indicators: Blue sticky notes contain a specific, accurate claim that references either xylem or phloem and at least one of: direction, substance transported, or cell type. Orange sticky notes show genuine intellectual curiosity about mechanism (force, energy, why) rather than definitional questions ('what does phloem mean?'), indicating students are ready for the mechanistic lessons ahead. Teacher photographs the DQB at end of lesson for the ARES platform record.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | showing progress in the evidence-gathering sequence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | highway against gravity?'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a><br/> Source: Seavuria<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Vascular Tissue of Plants - Advanced (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Students open a dedicated 'Plant Transport Model' page in their science notebooks (or use the printed model template provided on the ARES platform). They draw or paste a simplified outline of a whole maize plant. Using the colour conventions established today (blue = xylem pathway, red = phloem pathway), they: (1) draw the two vascular tissues as parallel lines running from root to leaf tip; (2) label what each tissue carries; (3) draw directional arrows; (4) add one annotation box per tissue explaining one structural feature (e.g., 'xylem = dead hollow cells, no nucleus, form continuous tubes'); (5) draw a thought-bubble next to the wilting leaf with the current best explanation for wilting: 'Insufficient water reaching leaf cells via xylem → loss of turgor pressure → wilting.' Students label this 'MODEL v.1 — Lesson 3' and leave a large blank margin/space titled 'What I will add next lesson' — reinforcing that this is a living, growing model.</p> | <p>Say: 'The last 8 minutes belong to your model. This is the most important thing you will do today — not because it is perfect, but because YOU are building it and you will add to it every lesson until it answers the farmer's full puzzle.' Project the model template on the board. Walk through the 5 steps one at a time: 'Step 1 — draw your maize plant outline. 30 seconds.' [Wait.] 'Step 2 — draw two parallel lines from root to tip. Label them. 30 seconds.' [Continue for each step.] With 2 minutes remaining: 'In your thought bubble, write your best explanation for the wilting — use the words xylem and turgor pressure.' [Circulate and give specific affirmations: 'Kamau, your arrow direction is exactly right. Fatuma, I love that you added the maize cob as a phloem sink — well done.'] Close: 'Label this Model v.1. In Lesson 4 we will add the mechanism that pulls water UP through the xylem — and your model will become even more powerful.'</p> | <p>Strategy: Progressive Model Revision (Cumulative Science Notebook Model). By explicitly versioning the model (v.1, v.2...) students experience science as an iterative, evidence-driven process rather than a fixed body of facts. The blank 'What I will add next lesson' margin creates anticipatory scaffolding — students enter Lesson 4 already wondering what belongs in that space, priming attention and motivation.</p> | <p>Observable indicators: Model v.1 contains (a) correctly colour-coded xylem and phloem pathways; (b) directional arrows consistent with Phase 2 diagram work; (c) at least one accurate structural annotation per tissue; (d) a thought-bubble explanation that correctly links xylem, water, turgor pressure, and wilting in a causal chain — even if incomplete. Teacher uses a rapid 3-point holistic rubric (3 = all four elements present; 2 = three elements; 1 = fewer than three) during final circulation to identify students needing a check-in at the start of Lesson 4.</p> |

## D. TEACHER REFLECTION

After the lesson, consider the following reflection questions: (1) **DIAGRAM ACCURACY:** When I circulated during Phase 2, how many students had the phloem arrow direction wrong (downward only, missing the sink-directed upward flow to the maize cob)? What does this tell me about how I introduced bidirectional flow — did I give enough time for this to sink in, or did I rush the reveal? (2) **JIGSAW QUALITY:** During the jigsaw in Phase 3, did the 'xylem' pairs actually teach the passive/active energy distinction accurately, or did they skip it? If most pairs skipped row four of the Venn diagram, I need to re-address the energy requirement of xylem transport explicitly at the start of Lesson 4 before moving to transpiration pull. (3) **DQB ORANGE NOTES:** What proportion of students' new questions were mechanistic ('what force?' 'what happens inside the cell?') vs. definitional ('what does xylem mean?')? A high proportion of definitional questions signals that vocabulary load was too heavy and I should build in more explicit vocabulary scaffolding in Lesson 4. (4) **MODEL v.1 QUALITY:** Did students' thought bubbles show a genuine causal chain (xylem → water → turgor → wilting) or just a label ('wilting because no water')? Shallow thought bubbles indicate students can recall facts but cannot yet reason causally — a target for Lesson 4 questioning. (5) **TIMING:** Was 80 minutes sufficient for all five phases? If Phase 2 ran long (a common occurrence with diagram work), which phase was compressed, and how did that affect the quality of the DQB and Model phases? Plan to use a visible countdown timer in the next lesson.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>What did I observe?</b>                   | Students observed a whole-plant diagram of a maize plant (rooted in Kibwezi soil) showing two distinct vascular systems — xylem and phloem — running continuously from root tip to leaf tip and to the developing maize cob. They observed that blue (xylem) arrows point strictly upward, while red (phloem) arrows are bidirectional — primarily downward but also toward the cob as a sink. They observed from the reading that xylem is composed of dead, hollow cells requiring no living machinery to function, while phloem is made of living cells.                                                                                                                                                                                                                                                                               |
| <b>What did I learn?</b>                     | Students learned that plants have two specialised vascular transport tissues: (1) Xylem — transports water and dissolved mineral salts from roots upward to leaves; composed of dead, lignified cells that form continuous hollow tubes; transport is unidirectional (upward). (2) Phloem — transports manufactured food substances (sugars/assimilates) from leaves to all other parts of the plant including roots and storage organs (e.g., maize cob); composed of living sieve tube cells; transport is bidirectional (source to sink). Students learned to locate both tissues within root, stem, and leaf cross-sections and to represent their directional flow using a standardised colour-coding system.                                                                                                                        |
| <b>How does this explain the phenomenon?</b> | Students used their new knowledge of xylem and phloem to deepen their explanation of the Kibwezi wilting maize phenomenon: the xylem highway must continuously deliver water from the soil (absorbed by root hairs via osmosis, which Lesson 4 investigates) upward through the stem to the leaves. If this upward delivery is insufficient — whether due to high evaporative demand on a hot afternoon, a long path length in tall maize, or competition from the dry riverbed soil — the leaf cells lose turgor pressure and the plant wilts. This explains why the farmer's morning watering does not guarantee afternoon hydration of the topmost leaves. Students recorded this causal chain in Model v.1 and posted evidence claims on the Driving Question Board, advancing the class's collective answer to the driving question. |

## LESSON 4 (80 minutes (double lesson)): The Entry Point: Root Structure and Water Absorption

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | To explain how water and mineral salts enter the root of a plant, linking root structure to the mechanisms of osmosis and active transport.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Knowledge</b>            | Students will know: (1) the structural features of root hair cells and how they increase surface area for absorption; (2) the pathway of water from soil through root hair, cortex, endodermis, and into xylem; (3) that water enters by osmosis (passive) and mineral salts enter by active transport (requiring energy); (4) the role of water potential gradients in driving osmosis.                                                                                                                                                                                       |
| <b>Skills</b>               | Students will be able to: (1) prepare and observe a hand-lens drawing of a maize or bean root tip, labelling root hairs, cortex, endodermis, and xylem; (2) set up and record results from a potato-strip osmosis experiment; (3) construct a labelled diagram showing the pathway of water entry into the root; (4) use evidence from the osmosis experiment to support an explanation of wilting in Kibwezi.                                                                                                                                                                 |
| <b>Attitudes</b>            | Students will appreciate the precision of biological structures — that the microscopic root hair cell is the critical gateway determining whether a maize plant in Kibwezi survives or wilts, fostering wonder at the relationship between form and function.                                                                                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Purpose in Storyline</b> | Lessons 1–3 established what is happening in the Kibwezi maize plants (wilting despite watered soil) and introduced the vascular transport system at a macro scale. Lesson 4 zooms in to the entry point: the root. Students now gather direct observational and experimental evidence for HOW water actually crosses from soil into the plant — the osmosis mechanism — and why that mechanism can fail under dry, saline, or high-transpiration conditions. This evidence is the mechanistic foundation for understanding the full journey of water in Lessons 5–8.          |
| <b>Safety Notes</b>         | <ul style="list-style-type: none"> <li>– Handle scalpels/razor blades for cutting potato strips only under teacher supervision; always cut away from the body on a cutting board.</li> <li>• Wash hands after handling soil and roots.</li> <li>• If using iodine stain for root sections, avoid contact with skin and eyes — rinse immediately with water if contact occurs.</li> <li>• Salt solutions are non-toxic but label all beakers clearly to avoid confusion.</li> <li>• Dispose of potato strips and root material in the compost/bin — not in the sink.</li> </ul> |

### B. LESSON OVERVIEW

In this 80-minute double lesson, students first revisit the Kibwezi wilting puzzle with a targeted prediction: if water is in the soil, what is stopping it from entering the maize root? They then observe real maize or bean seedling roots (grown in the school garden) under hand lenses and, where possible, under a microscope, producing scientific drawings of root hair cells and the internal cross-section of the root. Next, they conduct a structured osmosis investigation using potato strips in tap water versus concentrated salt solution — a direct analogue for root-hair cells in soil versus salty soil — and record quantitative data (mass or length changes). In the Explain phase, students use their observations and data to construct a causal explanation: water moves by osmosis down a water-potential gradient from soil into root hair cells, then across the cortex and through the endodermis into the xylem, while mineral salts move against their concentration gradient via active transport requiring ATP. Students then update the class Driving Question Board (DQB) and add the 'Root Entry' module to their cumulative transport model, explicitly connecting their findings back to why the Kibwezi farmer's maize wilts even when she has watered.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Introduction to passive and active transport</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Active Transport (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Students receive a short written prompt (displayed on the board or ARES screen): 'The Kibwezi farmer watered her maize field at 7 a.m. By noon the plants are wilting. Water IS in the soil — you can see it. Write your best explanation: what do you think has to happen for that soil water to actually get INSIDE the root and travel upward? What might be stopping it?' Students write individual predictions in their science notebooks for 4 minutes, then share with a partner (Think-Pair-Share). Selected pairs share with the class. The teacher records key prediction words on the board in two columns: 'What helps water enter' and 'What might block water entry.' Students also revisit their cumulative model from Lesson 3 and annotate it with a question mark at the root zone, labelling their prediction.</p> | <p>Display the prompt and say: 'Before we touch any equipment today, I want you to think hard about the root — the entry point. Look at your model from last lesson. There is a gap here at the root. Today we fill it with real evidence.' Allow 4 minutes of silent individual writing. Circulate and note 3–4 interesting or contrasting predictions without commenting. After pair discussion (2 minutes), cold-call exactly 4 pairs: 'Pair 1 — Akinyi and Mwangi — what did you predict? Pair 2 — Fatuma and Otieno — do you agree or disagree, and why?' Record responses verbatim on the board. Explicitly say: 'I am NOT telling you if you are right. We are going to use evidence to test these ideas.' WAIT TIME: after each pair responds, wait 10 seconds before calling the next pair to allow class to process. Circle the word 'osmosis' if any student uses it; if none do, write it on the board with a question mark and say: 'Some biologists use this word to describe water movement — keep it in mind as we observe today.'</p> | <p>Think-Pair-Share Prediction Elicitation — by externalising prior conceptions before observation, students create a cognitive 'gap' between what they think and what the evidence will show. Recording predictions publicly on the board creates accountability and makes conceptual change visible at the end of the lesson when students revisit the board.</p>                                                                                                               | <p>Teacher scans notebook predictions during circulation. Look for: (1) Does the student invoke any mechanism (pressure, sucking, diffusion, osmosis) or only describe the outcome? (2) Does the student connect root structure to function? Students who only write 'the water goes up through the roots' need targeted questioning during the Observe phase. Collect one notebook prediction snapshot mentally from a struggling learner to scaffold during Phase 2.</p> |
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| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Osmosis</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Active transport</a><br/> Source: Khan Academy (English - US curriculum)</p>                                                                                                                                              | <p>PART A — Root Observation (25 minutes): In groups of 4, students receive a tray containing: 2–3 maize or bean seedlings (7–10 days old, grown in the school garden or in soil-filled cups), a hand lens, a ruler, and a prepared worksheet with a blank circle for a scientific drawing. Students: (1) gently remove one seedling from soil and rinse the root under a small cup of water; (2) use the hand lens to observe root hairs — they sketch and label what they</p>                                                                                                                                                                                                                                                                                                                                                          | <p>Before Part A begins, hold up a seedling and say: 'This maize seedling was growing in our school garden — the same species as the Kibwezi farmer's crop. You are going to look at the part of the plant that is SUPPOSED to bring in water. Look carefully. What do you notice?' WAIT 10 seconds. Circulate during drawing; prompt with: 'How many root hairs can you count on 1 cm of root? Why might having MORE root hairs matter for the Kibwezi farmer's maize?' For</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Structured Observation with Annotated Scientific Drawing — requiring students to produce a labelled drawing (not just look) forces active processing of structural features. Connecting the potato strip experiment directly to the Kibwezi phenomenon ('which strip feels like the wilted maize?') prevents the investigation from becoming an isolated activity; it is always evidence for the driving question. The shared class data table on the board makes patterns</p> | <p>Circulate and check drawings: every drawing must include at least 3 labels. If a student labels only 'root hair' and nothing else, prompt: 'What is INSIDE the root behind the hair? Can you find it on the cross-section?' For the osmosis experiment, listen for causal language: a student who says 'the salt one got shorter because salt took the water' is beginning to apply osmosis correctly. A student who says 'the salt one dried out' has an</p>           |

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| <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                           | <p>see, measuring the approximate length of root hairs in mm; (3) where a microscope is available, the teacher (or a trained student) has pre-prepared a thin cross-section of a root stained with dilute iodine, mounted on a slide — groups rotate to view it and add to their drawings, labelling root hair cell, cortex cells, endodermis, and central xylem. Where no microscope is available, students use a printed ARES diagram of a root cross-section and annotate it. PART B — Osmosis Investigation (20 minutes): Each group receives 4 potato strips of equal length (approx. 5 cm) cut from a raw Irish potato. Two strips go into a beaker of tap water; two go into a beaker of concentrated salt solution (2 tablespoons of table salt per 250 ml water — a proxy for dry, salty soil). Students measure and record initial length/mass of each strip, leave for 15 minutes while completing Part A drawings, then measure final length/mass and record observations (turgid vs. flaccid, firm vs. floppy). Students record all data in a results table in their notebooks.</p> | <p>microscope rotation, say: 'You have 90 seconds at the microscope. Focus on finding the xylem in the centre. Point to it on your drawing.' For Part B setup, demonstrate cutting one potato strip and say: 'All strips must be the same length — this is our controlled variable. Why?' Cold-call 2 students to answer. After 15 minutes, prompt groups: 'Before you measure, PREDICT: which strips will be longer — water or salt solution? Write your prediction NOW.' Then have them measure and record. Ask: 'Which strip feels like the maize stem on a good morning? Which feels like the wilted maize at noon in Kibwezi?' WAIT 12 seconds. Cold-call 3 groups to report their results aloud; record class data in a shared table on the board.</p> | <p>visible across groups, prompting discussion of variability and reliability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>incomplete model — note this for the Explain phase. Collect class data table results — if more than 2 groups show unexpected results (e.g., salt strip got longer), pause and troubleshoot together before moving on.</p>                                                                                                                                                                                                                                                    |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Osmosis</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>HTML:</b><br/> <a href="#">Active Transport (Flexbook)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Students first complete a Claim-Evidence-Reasoning (CER) frame in their notebooks: CLAIM: 'Water enters the root by...' EVIDENCE: (cite their potato strip data and root hair observation) REASONING: (explain WHY the evidence supports the claim, using the concept of water potential / concentration gradient). The teacher then leads a whole-class explanation using a large diagram of a root cross-section drawn on the board, annotating it live as students contribute. Key concepts</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Start the Explain phase by saying: 'You now have two pieces of evidence — your root drawing and your potato strip data. A scientist does not just describe what happened; they explain WHY. Let's build that explanation together.' Write on the board: WATER POTENTIAL — say 'This is the key idea. Water always moves from where it has MORE potential energy to where it has LESS — like water flowing downhill. In Kibwezi, we need to ask: is the water in the soil at</p>                                                                                                                                                                                                                                                                           | <p>Claim-Evidence-Reasoning (CER) Framework — CER requires students to explicitly link their empirical data (potato strip results, root hair observations) to conceptual claims (osmosis, active transport), and to articulate the logical reasoning that connects them. This combats the common error of treating the experiment as separate from the biology concept. Live annotated diagram building ensures the explanation is co-constructed, not just transmitted, increasing retention and</p> | <p>Collect CER notebooks from 6 students (2 high, 2 middle, 2 lower based on Phase 1 assessment) and scan during Phase 4 DQB work. Indicators of mastery: (1) Claim uses the word 'osmosis' correctly; (2) Evidence cites specific data ('the strip in salt solution decreased in length by 1.2 cm'); (3) Reasoning mentions 'water potential' or 'concentration gradient' and direction of movement. Indicator of misconception: 'Water was absorbed by osmosis AND active</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                       | <p>built: (1) water potential gradient — soil water (high) → root hair cell (lower) → cortex (lower) → xylem (lowest) drives osmosis passively; (2) the Casparian strip in the endodermis forces all water through the living cells (symplastic pathway) — a checkpoint; (3) mineral salts move against their concentration gradient into root hair cells via active transport — this requires ATP from root cell respiration; (4) connection back to Kibwezi: in dry, salty, or compacted soil, the water potential of the soil may be LOWER than the root hair cell, reversing or stopping osmosis — this is why the farmer's maize wilts even with some soil water present if the soil salinity is high or the root hairs are damaged. Students then return to their predictions from Phase 1 and annotate them: 'My prediction was ✓ correct / ✗ incorrect / ~ partially correct because...'</p> | <p>higher or lower potential than inside the root hair cell?' Cold-call 3 students for responses. WAIT 10 seconds each time. As students contribute, draw arrows on the root diagram showing the direction of water movement. When you reach the endodermis, say: 'Here is something surprising — there is a waxy barrier here called the Casparian strip. It FORCES water to go through a living cell. Why might the plant WANT to control what gets in here?' WAIT 12 seconds. Cold-call 2 students. For mineral salts, ask: 'If mineral salts are in low concentration in soil and the plant needs them in high concentration inside — can they move by osmosis? No — they move by active transport. This costs the plant energy. Where does that energy come from?' Accept 'respiration' and elaborate. Conclude by returning to Kibwezi: 'If the farmer's soil becomes very salty — perhaps from poor-quality irrigation water — what happens to the water potential of the soil? Does water enter the root or leave it?' WAIT 15 seconds. Cold-call 2 students. Summarise on the board: OSMOSIS (passive, water potential gradient) + ACTIVE TRANSPORT (energy, mineral salts) = ROOT ABSORPTION.</p> | <p>ownership.</p>                                                                                                                                                                                                                                                                                                                            | <p>transport' — the student has confused the mechanisms for water vs. mineral salts. Address this in Phase 4 discussion.</p>                                                                                                                                                                                                                                                       |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Introduction to passive and active transport</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a></p> | <p>Students return to the class Driving Question Board. Working first individually (2 minutes) then as a class (5 minutes), students: (1) identify which questions on the DQB they can now ANSWER with today's evidence — they write answers on green sticky notes and add them to the board; (2) identify NEW questions today's lesson has raised — written on yellow sticky</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Point to the DQB and say: 'In Lesson 1 we put this question up: Why does the Kibwezi maize wilt even though water is in the soil? We now have real evidence. Turn to your partner — can you answer ANY part of this question using what you found today? You have 2 minutes.' After pair discussion, cold-call 3 students to place their green sticky notes on the board</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Driving Question Board Iterative Update — the DQB makes the epistemological progress of the class visible and cumulative. By distinguishing 'answered' (green) from 'new questions raised' (yellow), students experience science as a process of progressively deeper understanding rather than a checklist of facts. Anchoring every</p> | <p>Read green sticky notes posted: check that students are citing mechanisms (osmosis, water potential, active transport) not just restating the phenomenon ('water goes in'). A sticky note that reads 'Water enters by osmosis because the root hair cell has lower water potential than the soil' scores at mastery. A note that reads 'Water enters through root hairs' is</p> |

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| <p><a href="#">for similar videos</a></p> <p> <b>EXERCISE:</b><br/><b>Active transport</b><br/>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>notes; (3) revise any predictions or partial answers from Lessons 1–3 that today's evidence updates. Specifically, students address: 'Why does the Kibwezi farmer's maize wilt even when soil has water?' — students should now be able to give a partial mechanistic answer: 'If soil water potential is lower than root hair cell water potential (e.g. due to salt or dry compacted soil), water cannot enter by osmosis, so the plant wilts despite water being present.' Students also address: 'Why do plants near the dry riverbed wilt sooner?' — root hairs in drier, more saline sandy soil near the riverbed may face unfavourable water potential gradients earlier.</p>                                                                                                                                                                                                                              | <p>and read their answers aloud. For each answer, ask the class: 'Does everyone agree? Is there anything missing from this explanation?' WAIT 10 seconds. Then ask: 'What NEW questions has today's lesson given you? What do we STILL not know?' Prompt if needed: 'We know water enters the root — but how does it travel UP to a leaf 2 metres high in a maize plant? Does it just... sit there? Add that question if it's not already on the board.' Ensure the board visually tracks the progression: Lessons 1–3 evidence cluster vs. today's new evidence cluster.</p>                                                                                                                                                                                                                                                                                                                                          | <p>DQB update to the Kibwezi phenomenon ensures the board remains a living narrative, not a vocabulary list.</p>                                                                                                                                                                                                                                                                                                                                                        | <p>descriptive only — prompt that student: 'Can you add HOW it enters and WHY it moves that direction?' Count the number of new yellow-note questions — a rich set of questions (e.g. 'How does water go up against gravity?', 'What happens to root hairs in very dry soil?') indicates high cognitive engagement.</p>                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/><a href="#">12 Plant Transport Xylem And Phloem</a>, <a href="#">Transpiration [3D Animation]</a><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: Seavuria<br/> <a href="#">Search ARES</a><br/><a href="#">for similar videos</a></p> <p> <b>EXERCISE:</b><br/><b>Active Transport Practice</b><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: CK-12<br/> <a href="#">Search ARES</a><br/><a href="#">for similar readings</a></p> | <p>Students retrieve their cumulative transport model (started in Lesson 1 and updated each lesson). Today they add a detailed 'Root Entry Module' to the model. Using a template provided on ARES (or drawn freehand), they draw and annotate: (1) soil particles with water surrounding them (labelled 'high water potential'); (2) a root hair cell with its elongated extension, labelled with a lower water potential; (3) arrows showing water moving by OSMOSIS from soil → root hair → cortex cells → endodermis (with Casparian strip labelled) → xylem; (4) a separate arrow showing mineral ions entering by ACTIVE TRANSPORT with an ATP symbol; (5) a 'Kibwezi failure zone' annotation — a red dotted arrow showing what happens when soil water potential drops below root hair cell water potential (water moves OUT — the plant wilts). Students also write a one-sentence 'model statement' at</p> | <p>Distribute or display the model template. Say: 'Your model is your running scientific explanation of the Kibwezi puzzle. Every lesson, it gets more complete. Today you add the entry point — the root. This must connect to what is already on your model from Lessons 1–3.' Circulate and check that: (1) arrows for osmosis and active transport are in the CORRECT direction; (2) the Casparian strip is included; (3) the 'Kibwezi failure zone' annotation is present. If a student omits the Casparian strip, say: 'What forces water to pass through a living cell and not around it? You observed that structure today — add it.' Cold-call 2 students to present their model to the class in the final 3 minutes, specifically asking them to 'point to where the Kibwezi farmer's problem starts.' Assign the model as a take-home completion task if not finished in class. Preview Lesson 5: 'Your</p> | <p>Cumulative Annotated Model Revision — the running model functions as an external cognitive artefact that represents the student's growing mechanistic understanding. By requiring a 'Kibwezi failure zone' annotation, the teacher ensures the model is always explanatory (it must account for the phenomenon) not merely illustrative. The one-sentence 'model statement' forces students to synthesise visual and verbal representations, deepening encoding.</p> | <p>Exit check — before students leave, teacher does a rapid 'model gallery walk' of 6–8 open notebooks on desks. Check for: (1) Osmosis arrows pointing in the correct direction (soil → xylem); (2) ATP symbol present on active transport arrow; (3) 'Kibwezi failure zone' present with a direction-reversal annotation. Any model missing all three of these is flagged for a 5-minute catch-up conversation at the start of Lesson 5. Optionally, pose a verbal exit question to the whole class: 'In one word — what is the mechanism that pulls water from the soil into the root hair?' (Answer: osmosis). Cold-call 4 students for a rapid-fire round.</p> |

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|  | the bottom of their model page: 'Water enters the root by osmosis along a water potential gradient, while mineral salts enter by active transport; in Kibwezi's dry season, falling soil water potential can reverse or stop osmosis, causing the maize to wilt.' | model now shows water IN the xylem at the bottom of the root. Next lesson, we ask: how does it travel up — against gravity — to a leaf three metres high in a tall maize plant? Start thinking about that question tonight.' |  |  |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) OSMOSIS MISCONCEPTION CHECK — Did students correctly distinguish osmosis (water, passive, water potential gradient) from active transport (mineral ions, energy-requiring)? If more than a third of CER frames conflated the two, plan a 10-minute misconception-busting opener for Lesson 5 using the analogy of 'water sliding downhill freely vs. a person climbing uphill carrying a heavy load of minerals — the climber needs energy.' (2) POTATO STRIP DATA QUALITY — Did groups obtain consistent, measurable results (salt strip shorter/flaccid; water strip longer/turgid)? If results were inconsistent, evaluate whether salt concentration was too low or soaking time was insufficient. Adjust the protocol for future classes: increase salt to 3 tablespoons per 250 ml and extend soaking to 20 minutes. (3) DQB PROGRESSION — Count the green vs. yellow sticky notes. If yellow (new questions) outnumber green (answered questions) by more than 2:1, students may feel overwhelmed rather than progressive — validate this by framing new questions as 'scientific curiosity' at the start of Lesson 5. (4) KIBWEZI CONNECTION — Were students able to spontaneously apply today's mechanisms to the wilting phenomenon without being prompted? If not, consider adding a 'Kibwezi connector' card to Phase 3 with a sentence starter: 'This explains the Kibwezi farmer's problem because...' (5) EQUITY CHECK — Were students without microscope access equally engaged? If the printed ARES cross-section diagram was insufficient, explore whether a short 90-second video clip of a root hair cell under microscopy (downloadable for offline use via ARES) would bridge the gap for the next cohort.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>What did I observe?</b>                   | Students observed real maize/bean seedling roots under hand lenses and (where available) a microscope, producing labelled scientific drawings of root hair cells, cortex, endodermis, and xylem. They also observed potato strips in tap water vs. concentrated salt solution, recording changes in length/firmness after 15 minutes.                                                                                                                                       |
| <b>What did I learn?</b>                     | Students learned that water enters the root by osmosis — moving passively from soil (high water potential) through root hair cells, cortex, and endodermis into the xylem (low water potential). Mineral salts enter by active transport against their concentration gradient, requiring ATP. The Casparian strip in the endodermis forces all water through living cells, acting as a selective checkpoint. Root hairs increase surface area to maximise absorption rate.  |
| <b>How does this explain the phenomenon?</b> | Students explained that the Kibwezi farmer's maize wilts despite soil water being present because, in dry or saline soil, the water potential of the soil drops below that of the root hair cells — reversing or halting osmosis so water cannot enter (or even leaves the root). Plants near the dry riverbed wilt sooner because sandy, drier, more saline soils near the riverbed reach this critical low water potential threshold faster than the wetter lower slopes. |

## LESSON 5 (80 minutes): The Highway: Stem Structure and Vascular Bundles

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Purpose</b>              | To investigate the internal structure of herbaceous stems, identify xylem and phloem within vascular bundles, and connect stem anatomy to the movement of water and food through the plant — advancing students' understanding of why the Kibwezi maize wilts even after watering.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Knowledge</b>            | Students identify xylem and phloem within vascular bundles of herbaceous stems, describe their arrangement and structural differences, and distinguish between herbaceous and woody stem organisation.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Skills</b>               | Students prepare hand cross-sections of fresh stems using a razor blade, apply eosin or iodine stain, observe under a hand lens or microscope, draw and label vascular bundle diagrams, and set up the coloured water/celery experiment for Lesson 6.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Attitudes</b>            | Students appreciate the precision required in biological investigation, demonstrate care and patience when handling sharp instruments, and show curiosity about how microscopic structures solve the large-scale problem of moving water to the highest leaf of a maize plant.                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants? How are manufactured food substances transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | In Lessons 1–4, students discovered that water enters roots by osmosis, moves through root hair cells, and is pulled upward by transpiration pull. A critical gap remained: what physical structure inside the stem actually forms the 'highway' that carries water upward and sugars downward? Lesson 5 fills that gap by revealing the vascular bundle — the anatomical core of the transport system. Students now see the tubes, not just the forces, and can begin to explain why a damaged or blocked stem (drought stress, disease, or waterlogging causing anaerobic root death) would break the highway and cause the Kibwezi maize to wilt irreversibly. |
| <b>Safety Notes</b>         | Razor blades are extremely sharp. Demonstrate the correct cutting technique before students begin: hold the stem firmly on a cutting tile, curl fingers away from the blade, and use a single smooth downward stroke. Students must not rush. Collect all blades at the end of the activity and store them in a labelled container. Eosin dye can stain skin and clothing — students should wear aprons if available and wash hands thoroughly after the activity. Iodine is a mild irritant; avoid contact with eyes.                                                                                                                                            |

### B. LESSON OVERVIEW

Lesson 5 is the anatomical investigation lesson in the eight-lesson evidence-gathering sequence. Students move from understanding the forces that drive transport (osmosis, transpiration pull — Lessons 1–4) to examining the physical structures through which transport occurs. Working in groups of three, students use razor blades to cut thin cross-sections of fresh bean or sunflower stems collected from the school garden, stain the sections with eosin or iodine solution, and observe vascular bundle arrangement with hand lenses or compound microscopes. They draw and label xylem and phloem, compare their findings to a herbaceous vs. woody stem diagram, and update their cumulative class transport model. At the end of the lesson, each group sets up the coloured celery or white carnation experiment (results observed in Lesson 6), making a written prediction about which tissues will be stained. The lesson is anchored throughout in the Kibwezi phenomenon: if the vascular bundles are the highway, what would block or destroy that highway and cause irreversible wilting?

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES for readings</a><br/> Search terms:<br/> plant transport<br/> vascular bundle</p> | <p>Students are shown two items side-by-side: a whole fresh bean stem (from the school garden) and a photograph of a wilting Kibwezi maize plant. The teacher poses the Predict prompt: 'We know water travels up the stem from root to leaf. But if you sliced this stem across — like cutting a sugarcane stalk — what do you think you would see inside? Draw a quick cross-section diagram in your notebook and label anything you predict you would find. Then write one sentence: how do you think the inside of this stem connects to the wilting maize in Kibwezi?' Students work individually for 5 minutes, then do a 'Turn and Compare' with a partner, noting similarities and differences in their predictions. The teacher cold-calls 4 pairs (8 students) to share predictions and records key ideas on the Driving Question Board under the column 'What we PREDICT about the stem highway.' Common predictions (hollow tube, solid green, a single big pipe, many small pipes) are recorded without correction — all are treated as hypotheses to be tested.</p> | <p>1. Hold up the fresh bean stem and the Kibwezi photograph together. Say: 'In our last lessons we discovered the FORCES that pull water up — osmosis, transpiration pull. Today we ask: what is the STRUCTURE, the actual physical highway, inside this stem?' 2. Display the Predict prompt on the board. Say: 'Take 5 minutes — draw, don't worry about being right. This is a hypothesis.' Set a visible 5-minute timer. 3. After 5 minutes say: 'Turn to your partner. Compare your diagrams. What did you agree on? What is different?' Allow 2 minutes. 4. Cold-call 4 pairs. For each, ask: 'Tell us your prediction and your reasoning.' WAIT TIME: 12 seconds after asking before accepting answers. 5. Record all predictions on the DQB. Say: 'None of these are wrong yet — they are all hypotheses. Our job today is to gather evidence to test them.'</p> | <p>Predict-Observe-Explain (POE) — Activate prior knowledge and surface misconceptions before observation. Recording predictions publicly on the DQB creates accountability: students will return to their predictions after the dissection and must explain what changed and why. This strategy builds the habit of treating prior ideas as testable claims rather than fixed truths.</p>                                                                             | <p>Observable indicator: Every student has a drawn cross-section diagram with at least one labelled prediction in their notebook before the dissection begins. The teacher scans notebooks during the Turn-and-Compare phase. Students who have drawn only a blank circle or written 'I don't know' are prompted with: 'Think about how water travels in a pipe at home — is there one big pipe or many small ones? Draw what you think.'</p>                            |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES for similar videos</a></p>                                                                                                                                                                                                                                                                                                                                                                                  | <p>Groups of three receive: one fresh bean or sunflower stem (10–12 cm), a razor blade, a cutting tile, a petri dish or bottle cap with eosin or iodine solution, a hand lens (and compound microscope if available), a slide and coverslip, and a printed diagram showing herbaceous vs. woody stem structure for reference (not revealed until after initial observation). Step 1 (8 min): Each group cuts 2–3 thin cross-sections of their stem. The thinnest section</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>1. Distribute materials. Demonstrate the cutting technique FIRST with your own stem at the front: 'Watch carefully — I hold the stem flat, fingers curled, one smooth stroke downward. Now you try.' Allow one practice cut per group on a spare stem piece before cutting the real sample. 2. Circulate during staining and observation. Ask at each table: 'What colours do you see? Where are the coloured patches concentrated? What shape are</p>                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Observation-Inference Separation — A structured two-column recording tool ('What I See' vs. 'What I Think It Is') trains students in the NGSS Practice of Distinguishing between Data and Interpretation. This prevents students from drawing diagrams of what they expect rather than what they observe — a common error in biological drawing tasks. The subsequent comparison to the printed reference diagram is used as a self-correction tool, not as the</p> | <p>Observable indicators: (1) Each student's notebook contains a hand-drawn cross-section diagram with a minimum of four labelled structures, using two different pen colours to distinguish 'observed' from 'inferred/confirmed.' (2) Every group has a celery stalk set up in food dye with a written prediction recorded before the end of Phase 2. The teacher collects one notebook per group to check diagram quality and returns it before Phase 3. Red flag:</p> |

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| <p> <b>READING:</b><br/>No closely matching reading in the ARES library for this activity.</p> <p> <a href="#">Search ARES for readings</a><br/>Search terms:<br/>plant transport<br/>vascular bundle</p> | <p>is placed in the petri dish with staining solution for 3–4 minutes. Step 2 (5 min): The stained section is placed on a slide with a drop of water, covered, and observed under the hand lens or low-power microscope objective. Students observe and describe in writing: colour, shape, arrangement of any distinct regions or bundles they can see. Step 3 (10 min): Students draw a large, detailed cross-section diagram in their notebooks (minimum 8 cm diameter), labelling everything they can actually see — they must distinguish between 'I can see this' and 'I am inferring this.' Step 4 (5 min): The teacher reveals the printed herbaceous stem diagram. Students add correct labels (epidermis, cortex, vascular bundle, xylem, phloem, cambium, pith) to their own drawing using a different coloured pen, noting where their predictions were correct and where they were surprised. The coloured celery stalks (each group receives one celery stalk split at the base into two, each half placed in a different colour of food dye) are set up in bottles on the windowsill. Students write a prediction: 'In Lesson 6, I predict we will see colour in _____ because _____.'</p> <p>[KENYA CONTEXT: The teacher notes that the vascular bundles in a maize stem are scattered — different from the ring arrangement in bean stems — and draws a quick comparison: 'Think of a maize ugali — it comes from a monocot. Bean stew comes from a dicot. Even their stems are built differently!']</p> | <p>they?' WAIT TIME: 10 seconds after each question before offering a hint. 3. Prompt the distinction: 'Write two columns — WHAT I SEE (observation) and WHAT I THINK IT IS (inference). This is what real biologists like Dr. Miriam Khamala at the Kenya Plant Health Inspectorate Service do when they examine plant tissue.' 4. When revealing the printed diagram, say: 'Using a different colour pen, add the correct labels. Circle anything that surprises you compared to your prediction.' 5. For the celery setup, say: 'This is your evidence-gathering experiment for next lesson. Write your prediction NOW, in a full sentence with a reason. Do not change it once you've written it.' 6. Cold-call 3 students to read their celery predictions aloud. WAIT TIME: 12 seconds after asking. Record predictions on the DQB.</p> | <p>primary source of knowledge.</p> | <p>diagrams copied directly from the printed reference without a personal observation drawing — teacher intervenes and asks the student to identify one specific thing they personally saw through the hand lens.</p> |
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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport<br/> vascular bundle</p> | <p>The teacher leads a structured whole-class Jigsaw-to-Gallery discussion. Each group is assigned one of four 'expert' roles for 8 minutes: Group A — Xylem experts (structure, wall thickness, dead cells, role in water transport); Group B — Phloem experts (structure, sieve tubes, companion cells, role in sugar transport); Group C — Vascular Bundle Arrangement experts (ring in dicots vs. scattered in monocots, comparison of bean and maize); Group D — Connection-to-Phenomenon experts (how does the vascular bundle highway explain the Kibwezi wilting? What could block or destroy the highway?). Each expert group produces a 'mini-poster' (A4 paper, shared quickly at the front). Groups then regroup into mixed teams of four (one expert from each group) and teach each other for 8 minutes. Finally, the whole class comes together for a 4-minute Connect-Back: the teacher asks, 'Group D — what did you conclude? What could break the highway?' Cold-calls 3 students from different groups. Students individually write a two-sentence Explain statement: 'Water travels through _____ because _____. In Kibwezi, the maize might wilt permanently if _____ because _____.'</p> | <p>1. Assign expert roles clearly. Say: 'You have 8 minutes as experts. Your job is to understand your structure or concept well enough to TEACH it. Use your diagram, the printed reference, and your textbook.' Set a timer. 2. Circulate during expert group work. For Group D (Phenomenon group), prompt: 'What would happen if the xylem tubes got blocked by air bubbles — what do we call that? What about if the roots rotted from waterlogging? Would the wilting be reversible or not?' WAIT TIME: 12 seconds. 3. Signal regrouping: 'Move to your mixed groups. Each expert has exactly 2 minutes to teach — I will time you.' 4. Whole-class connect-back: 'Group D experts — in one sentence, how does today's anatomy lesson help explain the Kibwezi phenomenon?' Cold-call 3 students. WAIT TIME: 12 seconds each. 5. After each student speaks, ask the class: 'Does anyone want to add or challenge that?' WAIT TIME: 10 seconds. 6. Say: 'Now write your own two-sentence Explain statement. This is individual — no copying from partners.'</p> | <p>Jigsaw Cooperative Learning with Phenomenon Re-anchoring — The jigsaw structure ensures every student develops depth in one area and breadth across all areas, reflecting the NGSS Practice of Obtaining, Evaluating, and Communicating Information. The mandatory Group D (Phenomenon group) ensures that anatomical knowledge is not learned in isolation but is immediately connected back to the explanatory purpose: understanding the Kibwezi maize failure. The two-sentence Explain statement is a low-stakes writing-to-learn tool that makes thinking visible.</p> | <p>Observable indicators: (1) The two-sentence Explain statement correctly names xylem as the water pathway and phloem as the sugar pathway, and gives a plausible mechanistic reason for irreversible wilting (e.g., permanent xylem blockage by air embolism, cavitation, or root destruction). (2) During the whole-class Connect-Back, the teacher listens for whether students use causal language ('because,' 'which means that,' 'this causes') rather than purely descriptive language ('xylem is a tube'). Students using only descriptive language are prompted: 'That's what it is — but WHY does that structure matter for the wilting maize?'</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>The class returns to the whole-group Driving Question Board. The teacher reads aloud the two main KICD Key Inquiry Questions — 'How are water and mineral salts transported in plants?' and 'How are manufactured food substances transported in plants?' — and the overarching driving question about</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>1. Point to the DQB. Say: 'Look at what we predicted in Phase 1. Which predictions were correct? Which were wrong? Let's update honestly.' Cross out or annotate predictions on the DQB in a visible colour. 2. Distribute sticky notes (or ask students to write in their lesson journals). Say: 'Write TWO</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Evidence-Question-Phenomenon Sorting — The three-column DQB structure (Evidence / Questions / Connections to Phenomenon) trains students in the NGSS Practice of Engaging in Argument from Evidence by requiring them to distinguish between what they now know (evidence), what they still</p>                                                                                                                                                                                                                                                                              | <p>Observable indicator: Every student has contributed at least one sticky note or journal entry to each DQB column before leaving. The teacher scans entries for specificity — vague entries like 'I learned about stems' are redirected with: 'Can you name a specific structure you saw today</p>                                                                                                                                                                                                                                                                                                                                                           |

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| <p><a href="#">Animation</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/> <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport<br/> vascular bundle</p>        | <p>the Kibwezi maize. Each student writes two sticky notes (or two entries on the class chart if sticky notes are unavailable): (1) One piece of NEW EVIDENCE gathered today that helps answer the driving question — must reference today's stem dissection or observation. (2) One NEW QUESTION that today's lesson raised — something they are still wondering about. The teacher reads several aloud and sorts them into three DQB columns: 'Evidence we now have,' 'Questions still open,' and 'Connections to the Kibwezi Phenomenon.' The teacher highlights the bridge to Lesson 6: 'Tomorrow we will check our celery experiment. The coloured water will have travelled through the stem overnight — and you will see with your own eyes which tissue carries the water. That will either confirm or challenge what you have concluded today.'</p> | <p>things — one piece of evidence, one new question. You have 3 minutes.' Set timer. 3. Cold-call 4 students to share their evidence statements. WAIT TIME: 10 seconds after each. Ask: 'Is this evidence or inference? How do you know?' 4. Cold-call 3 students to share their new questions. Post them in 'Questions still open.' Say: 'Good scientists always leave lessons with more questions than they started with — that is a sign of real understanding.' 5. Point to the celery stalks on the windowsill: 'These are working for us overnight. Write in your notebook: I predict that by tomorrow, the coloured water will be visible in _____ because _____.'</p> | <p>need to find out (questions), and how it all connects to the real-world case they are trying to explain. Public posting of questions normalises intellectual uncertainty and creates shared curiosity for Lesson 6.</p>                                                                                                                                                                                                                                                                                                                                                                                   | <p>and explain what it does for the plant?' The teacher photographs the updated DQB at the end of the lesson to track conceptual progression across the eight-lesson sequence.</p>                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem.</a><br/> <a href="#">Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this</i></p> | <p>Students retrieve their cumulative transport model — the annotated diagram they have been building since Lesson 1, showing the full journey of water from soil to leaf. In Lessons 1–4, the model already includes: root hair cells, osmosis arrows, the cortex pathway, and the xylem column in the root. Today, students add the stem layer: they draw and label a cross-section of the herbaceous stem on the side of their model, showing vascular bundles with xylem (inner, thick-walled, dead cells, coloured blue or pencil-shaded) and phloem (outer, thin-walled, living, coloured red or dotted). They draw arrows showing: water moving UP through xylem, sugars</p>                                                                                                                                                                          | <p>1. Say: 'Take out your cumulative model. You are now going to add the stem to your plant transport highway. Use today's observation to make this accurate — not a copy from the textbook, but from what YOU saw.' Allow 5 minutes for individual model updating. 2. Circulate and check: are xylem and phloem correctly positioned (xylem inner/adaxial, phloem outer/abaxial in most herbaceous dicots)? Prompt incorrect models: 'Look at your cross-section diagram — which bundle tissue was closest to the centre? Which was closest to the outside?' WAIT TIME: 10 seconds. 3. Say: 'Now mark a DANGER ZONE — where do you think the highway is most</p>             | <p>Cumulative Model Revision — The running annotated diagram is the cornerstone sensemaking artefact of the eight-lesson sequence, embodying the NGSS Practice of Developing and Using Models. Each lesson adds one layer of mechanistic detail, so students can see — literally, visually — how their understanding has grown. The 'DANGER ZONE' annotation forces students to apply their new anatomical knowledge predictively, bridging from structure to function to the real-world failure case. Comparing models in small groups surfaces persistent misconceptions before they are consolidated.</p> | <p>Observable indicators: (1) The cumulative model now shows at minimum: root hair → cortex → xylem (root) → xylem (stem vascular bundle) → upward arrow to leaf, AND phloem with a downward sugar-transport arrow. (2) The 'DANGER ZONE' annotation is accompanied by a mechanistic justification (e.g., 'If xylem in the stem gets air-locked during extreme drought, water cannot reach the leaves and the plant wilts permanently — like in Kibwezi'). Models missing the phloem layer or showing only one undifferentiated 'tube' are flagged for follow-up at the start of Lesson 6.</p> |

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| <p>activity.</p> <p><a href="#">Search ARES for readings</a></p> <p>Search terms:<br/>plant transport<br/>vascular bundle</p> | <p>moving DOWN through phloem. They annotate: 'Xylem = one-way upward highway for water and minerals.' 'Phloem = two-way food highway, mainly downward from leaves.' Students also mark a 'DANGER ZONE' on their model — where they think the highway is most vulnerable to failure in a drought-stressed Kibwezi maize plant — and write a one-sentence justification. Groups of three briefly compare models (2 minutes) and agree on any corrections before the lesson ends.</p> | <p>fragile? Write one sentence explaining your choice.' Allow 2 minutes. 4. Say: 'Compare with your group for 2 minutes. If you disagree on the danger zone, keep both versions — we will test this idea in Lessons 6 and 7.' 5. Close by pointing to the wilting maize photograph: 'This farmer in Kibwezi is still waiting for an answer. Our model is getting closer to explaining her problem. What do we still need to know?' Accept 2 quick verbal responses. WAIT TIME: 10 seconds each.</p> |  |  |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions to inform Lesson 6 planning:

1. SAFETY EXECUTION: Did all students follow the razor-blade cutting protocol correctly? Were there any near-misses or rule violations? If so, adjust the Lesson 6 celery observation setup to minimise further blade use.
2. OBSERVATION QUALITY: When you collected notebooks during Phase 2, did students distinguish between their own observations and the reference diagram labels? If more than one-third of the class copied directly from the printed diagram without a personal observational drawing, begin Lesson 6 with a 5-minute 'blind observation' exercise — hand lens observation before any reference material is provided.
3. MISCONCEPTIONS SURFACED: The most common misconception in this lesson is that xylem and phloem are 'side by side as two separate tubes running the full length of the stem' rather than organised into discrete vascular bundles. Did this arise in DQB entries or model drawings? If so, use the celery cross-section in Lesson 6 to explicitly re-address it.
4. PHENOMENON CONNECTION: Did Group D (Phenomenon experts) produce a mechanistically rich explanation of how vascular bundle damage causes irreversible wilting, or did they stay at the descriptive level ('the tubes get blocked')? If only descriptive, plan a Lesson 6 opening hook: 'A sugarcane farmer in Kisumu notices that after a long dry spell, even when rain returns, some plants never recover — but some do. Using what you know about xylem and phloem, predict which damage would be reversible and which would not — and why.'
5. CELERY PREDICTIONS: Review the written celery predictions collected at the end of the lesson. How many students correctly predicted that only xylem would be stained? How many predicted the whole stem or the phloem? Use this data to plan targeted questioning during Lesson 6 observations — cold-call students whose predictions were wrong first, so they can experience productive cognitive conflict when they see the actual result.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <p><b>What did I observe?</b></p> | <p>Students cut cross-sections of fresh bean or sunflower stems, stained them with eosin or iodine, and observed distinct coloured patches arranged in a ring pattern (in dicot stems). They noted that some regions had thicker walls and appeared more densely stained than others. They also set up split celery stalks in two different food-dye colours and observed the initial uptake of colour beginning at the cut base.</p> |
| <p><b>What did I learn?</b></p>   | <p>The stem contains vascular bundles made up of two distinct tissues: xylem (inner, thick-walled, dead cells that form hollow tubes for carrying water and minerals upward) and phloem (outer, thin-walled, living cells that carry dissolved sugars mainly downward</p>                                                                                                                                                             |

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|                                              | from leaves). In herbaceous dicot stems the bundles are arranged in a ring; in monocot stems (like maize) they are scattered. These bundles are the physical 'highway' that connects roots to leaves.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>How does this explain the phenomenon?</b> | Water and mineral salts absorbed from the soil by root hair cells travel upward through the xylem vessels in vascular bundles running through the stem to every leaf. Sugars manufactured in leaves by photosynthesis travel downward through the phloem to roots, growing tips, and storage organs. In the Kibwezi maize, extreme drought stress can cause air embolisms (cavitation) in the narrow xylem vessels, permanently breaking the water highway — which is why some plants never recover even after watering returns. Plants like the sukuma wiki shaded by the mango tree lose less water through transpiration, reducing the tension in their xylem and making cavitation far less likely. |

## LESSON 6 (80 minutes): The Destination: Leaf Structure and Transpiration

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To investigate leaf anatomy and the process of transpiration, and to explain how water loss through stomata drives the wilting of maize in Kibwezi.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) describe the internal and external structure of a leaf in relation to transport and water loss; (b) define transpiration and identify the structures through which it occurs; (c) explain factors that affect the rate of transpiration.                                                                                                                                                                                                                                                                                |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to: (a) prepare and observe a stomatal impression from a sukuma wiki or Ficus leaf using clear nail varnish; (b) record and interpret results of the coloured dye experiment from Lesson 5; (c) construct a labelled diagram of a leaf cross-section showing xylem, stomata, cuticle, and mesophyll.                                                                                                                                                                                                                            |
| <b>Attitudes</b>            | The learner appreciates how microscopic leaf structures have large-scale consequences for crop survival, and develops curiosity about how environmental factors — shade, wind, humidity — interact with plant physiology.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | How does water leave a plant, and why does the rate of this loss vary depending on conditions in places like Kibwezi?                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | In Lessons 1–5, students traced water from the Kibwezi soil into roots (osmosis), up the stem (xylem/transpiration pull), and confirmed xylem transport using coloured dye. Lesson 6 closes the journey — students now investigate the DESTINATION: the leaf. They discover that the same stomata that let CO <sub>2</sub> in for photosynthesis are the primary exit point for water, and that the rate of this loss (transpiration) is what determines whether the farmer's maize wilts or survives. This lesson makes the wilting phenomenon fully explicable for the first time. |
| <b>Safety Notes</b>         | Clear nail varnish contains acetone-based solvents — ensure the classroom or laboratory is well ventilated. Students must not apply nail varnish near open flames. Handle razor blades or scalpels (for leaf cross-sections) only under teacher supervision; pass blades with the handle facing the recipient. Wash hands after handling plant material. Students with known skin sensitivities should wear disposable gloves.                                                                                                                                                       |

### B. LESSON OVERVIEW

Students begin by reviewing the 'sweating leaf bag' demonstration set up on a branch in the school compound before the lesson — water droplets visible inside the bag provide immediate, concrete evidence of transpiration. They then record and interpret the coloured dye results from Lesson 5, confirming that xylem carries water from root to leaf. The main practical involves preparing a stomatal impression from a sukuma wiki leaf using clear nail varnish, peeling it off, and observing guard cells and stomata under a hand lens or microscope. Students dissect a fresh leaf to observe veins (vascular bundles), the waxy cuticle, and upper vs lower epidermis. Using all gathered evidence, they define transpiration, identify the factors that affect its rate, and explicitly link those factors to the Kibwezi afternoon conditions (high temperature, low humidity, wind). A structured class discussion uses a Claim–Evidence–Reasoning (CER) frame to answer: "Why does the maize wilt even though the soil was watered?" Students update their Driving Question Board and revise their cumulative transport models to include leaf anatomy and stomata.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience               | Teacher Moves                       | Sensemaking Strategy             | Formative Assessment Strategy     |
|----------------------|----------------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| <b>Predict Phase</b> | Students walk outside (or view a | Before class: tie a clear polythene | Think–Pair–Share with Prediction | Teacher scans written predictions |

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| <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> Source: Seavuria<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Photosynthesis And Transpiration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: iicba<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>photograph projected on screen) to observe the 'sweating leaf bag' — a clear polythene bag tied tightly around a leafy branch of the school's mango or Ficus tree before school started. They observe water droplets on the inside of the bag. Back in class, students write a 2-sentence individual prediction in their notebooks answering: 'Where did the water inside the bag come from, and how did it get there?' They then share predictions with a shoulder partner (Think–Pair–Share). Teacher selects 4 students using cold-call to share with the whole class — targeting students who gave contrasting predictions (e.g., one who says 'rain condensed' vs one who says 'the leaf sweated it out').</p> | <p>bag tightly around a leafy branch in the school compound. At the start of the lesson, say: 'Before we open our books, I want you to walk to the mango tree/Ficus outside — or look at this photo — and observe this bag carefully. Do NOT touch it yet. You have 90 seconds. Look, think, write one observation.' After students return, say: 'Now write TWO sentences: Where did that water come from, and how did it travel there?' Allow 2 minutes of silent individual writing. Then: 'Turn to your shoulder partner — share your prediction. You have 60 seconds each.' Cold-call 4 students — deliberately select 2 with 'condensation from air' type answers and 2 with 'plant released it' type answers. Record both prediction categories on the board under the heading: 'OUR PREDICTIONS — we will test these today.' Say: 'Keep your prediction written down — we will return to it at the end of the lesson and see who was right, and WHY.'</p> | <p>Anchoring. By writing predictions individually first, students commit to a prior model before peer influence. Recording contrasting predictions on the board creates productive cognitive dissonance — students are motivated to find out who is right. This directly connects to the Kibwezi phenomenon: if a bagged leaf produces water, the plant is actively losing water, not just sitting passively.</p>                                                                                                                                                                      | <p>during pair-share. Observable indicator of readiness: student predictions reference either 'the leaf' or 'the plant' as the source (even if mechanism is wrong). Red flag: students who write only 'I don't know' — teacher pairs these students with a confident peer and circulates to prompt: 'What do you think is INSIDE the leaf that could become water?'</p>                                                                                                                                                                                                                                          |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> Source: Seavuria<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Photosynthesis And Transpiration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: iicba</p>                                                                                                              | <p>Students carry out THREE interlocking observations in this phase:</p> <p>(A) DYE EXPERIMENT RESULTS (from Lesson 5): Students retrieve their cross-sections of celery stalks soaked in red/blue food dye. Working in pairs, they observe which tissue is stained, draw a labelled diagram, and write one sentence: 'The dye moved through _____ tissue, which tells us that _____ carries water up the stem.' Groups share findings briefly (2 minutes).</p>                                                                                                                                                                                                                                                        | <p>Distribute materials in pre-labelled trays: sukuma wiki leaves, clear nail varnish, clear tape, glass slides, hand lenses, razor blades in holders, printed reference cross-section diagrams, coloured pencils. Say: 'We have THREE observations to complete in the next 25 minutes. I will time each one. Listen carefully to instructions before touching any equipment.'</p> <p>For Observation A: 'Take out your dye stem from Lesson 5. In your notebook, draw what you see — which tissue is coloured? Write one sentence explaining what this</p>                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Triangulated Evidence Building: Students simultaneously hold three pieces of evidence — dye in xylem (Lesson 5 confirmation), stomata visible on leaf surface (new), and leaf cross-section showing vascular bundles connecting to mesophyll (new). This triangulation strategy — named explicitly for students ('We are triangulating: three pieces of evidence pointing to the same conclusion') — mirrors how scientists build confidence in a claim. Each observation alone is suggestive; together they build a coherent mechanistic story of water moving from xylem into</p> | <p>Teacher checks diagrams during circulation. Observable indicators: (1) Student correctly identifies stained tissue as xylem in observation A. (2) Student records MORE stomata on lower surface than upper in observation B — if equal or reversed, teacher asks 'Which side did you apply the varnish to?' (3) Student's cross-section diagram includes at least 4 correctly labelled structures. Red flag: students who cannot distinguish xylem from phloem in cross-section — teacher points to the vascular bundle and asks: 'Which of these two is closer to the upper surface? That is the xylem.'</p> |

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| <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                              | <p>(B) STOMATAL IMPRESSION PRACTICAL: Each student takes a fresh sukuma wiki leaf (or Ficus). They apply a thin layer of clear nail varnish to a 2 cm<sup>2</sup> patch on the LOWER surface of the leaf. After 2–3 minutes (while doing observation C), they peel the dried film off using a piece of clear adhesive tape, place it on a glass slide, and observe under a ×10 hand lens or the class microscope. Students draw what they see: guard cells, stomatal pore, epidermal cells. They repeat on the UPPER surface and compare stomata density. They record: 'Upper surface — number of stomata visible: _____. Lower surface — number of stomata visible: _____. Students also observe the leaf surface by touch and note the waxy, slightly shiny feel of the upper cuticle.</p> <p>(C) LEAF DISSECTION / CROSS-SECTION: Using a razor blade (teacher-supervised), students make a thin cross-section of the same sukuma wiki leaf midrib. They place it on a slide with a drop of water and observe under the hand lens. Students identify and label: upper epidermis, palisade mesophyll, spongy mesophyll, vascular bundle (xylem + phloem), lower epidermis, stomata. A printed reference diagram is provided for comparison.</p> | <p>tells us about water transport.' Allow 4 minutes. Cold-call 2 students: 'Tell me exactly which tissue is stained and what that means.'</p> <p>For Observation B: Model the nail varnish technique yourself first: 'Watch — I apply a thin layer here on the LOWER surface, like painting. Not thick — thin. We wait 2–3 minutes, then peel with tape. While we wait, we start Observation C.' After peeling: 'Place your film on the slide. What do you see? Count the pore-like openings on upper vs lower surface. Record your numbers.' Allow 10 minutes total. Circulate and ask: 'Why do you think there are more stomata on the lower surface? Discuss with your partner — 30 seconds.'</p> <p>For Observation C: 'With your blade FLAT and supervised by me, make a thin slice across the midrib. Water on slide. Observe. Use the reference diagram to help you label your drawing.' Allow 8 minutes. Say: 'I am looking for at least FIVE labelled structures in your diagram.' WAIT 10 seconds after each instruction before accepting questions.</p> | <p>mesophyll cells and out through stomata.</p>                                                                                                                                                                                   |                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">The water cycle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English)</p> | <p>Teacher introduces the formal definition of transpiration using the evidence students have just gathered. Students work in groups of 4 to complete a Claim–Evidence–Reasoning (CER) scaffold in their notebooks:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Open the Explain phase by returning to the board predictions: 'Let us go back to what you predicted. [Point to board.] Some of you said condensation from air. Some said the plant released it. Now you have evidence. Which is</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Claim–Evidence–Reasoning (CER) Framework: CER is named explicitly and its purpose explained — 'Scientists do not just state claims; they justify them with evidence and explain the logic connecting the two. We are doing</p> | <p>Collect CER notebooks from 6 randomly selected students (one per group) during the transition to the next phase. Observable indicators of proficiency: (1) Claim correctly names transpiration/stomata as source of</p> |

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| <p>- US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Photosynthesis And Transpiration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: iicba<br/> <a href="#">Search ARES</a><br/> for similar readings</p> | <p>CLAIM: 'The water inside the polythene bag came from _____.'</p> <p>EVIDENCE: Students list at least THREE pieces of evidence from today's observations (dye results, stomatal impressions, leaf bag water droplets).</p> <p>REASONING: Students write 3–4 sentences connecting their evidence to the claim using the terms: transpiration, stomata, xylem, guard cells, water vapour.</p> <p>Groups then tackle the Kibwezi question directly: 'It is 2:00 pm in Kibwezi. Temperature: 38°C. Wind: strong. Humidity: very low. Using what you now know about transpiration, explain in your group why the maize is wilting at this exact moment — even though the farmer watered the soil this morning.' Each group writes a group explanation (min. 5 sentences). Groups share out — teacher writes key phrases from each group on the board, building a class consensus explanation.</p> <p>Teacher then presents the FOUR factors affecting transpiration rate using a Kenya-relevant comparison table on the board:</p> <ul style="list-style-type: none"> <li>– Temperature (hot Kibwezi afternoon vs cool Kericho morning)</li> <li>– Humidity (dry Makueni vs humid Lake Victoria shore)</li> <li>– Wind speed (open maize field vs sheltered sukuma wiki under mango tree)</li> <li>– Light intensity (midday sun vs shaded plot)</li> </ul> | <p>supported? Turn to your group — 30 seconds — then I will cold-call.' Cold-call 3 students. Then say: 'The scientific term for what you are describing is TRANSPIRATION. Write this definition exactly: Transpiration is the loss of water vapour from a plant, mainly through the stomata of leaves.' Students write and underline.</p> <p>For the CER task: 'In your groups of four, complete the Claim–Evidence–Reasoning scaffold on page ____ of your workbook. You must cite at least THREE pieces of evidence from TODAY's observations. You have 8 minutes — go.' Circulate. At 5 minutes, say: 'You should be writing your reasoning now — use these words: transpiration, stomata, xylem, guard cells.' After 8 minutes, cold-call one group to share their full CER. Ask the class: 'Do you agree with their reasoning? What would you add?' Allow 2 student additions.</p> <p>For the Kibwezi group explanation: 'Now apply this to our farmer in Kibwezi. It is 2:00 pm. Hot. Windy. Dry. Explain exactly what is happening inside her maize plant RIGHT NOW.' Allow 6 minutes. Cold-call 2 groups. Construct the class consensus explanation on the board together. Say: 'This is the most important paragraph you will write in this unit — it answers our driving question for the first time completely.'</p> <p>For the factors table: present each factor with a Kenya contrast. Ask:</p> | <p>exactly what a plant physiologist at the University of Nairobi would do.' CER is particularly powerful here because students now have three independent pieces of evidence (dye, stomata, leaf bag) converging on one mechanism. The Kibwezi application task forces transfer — students must move from abstract definition to causal explanation of the anchoring phenomenon, which is the deepest level of sensemaking.</p> | <p>bag water. (2) Evidence section lists at least 3 specific observations (not generic statements). (3) Reasoning uses at least 3 of the 5 key vocabulary words correctly in context. (4) Kibwezi group explanation correctly identifies at least 2 factors (high temperature + low humidity OR high temperature + wind) as increasing transpiration rate beyond the rate of water uptake. Red flag for reteaching: students who explain wilting as 'the plant is thirsty' without referencing the rate mismatch between water uptake and water loss.</p> |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Students copy and annotate the table, adding arrows showing 'increases transpiration rate' or 'decreases transpiration rate' for each factor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'Why does the sukuma wiki under the mango tree barely wilt? [WAIT 12 seconds.] What factor has the mango tree changed?' Cold-call 2 students before explaining.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br>Source: Seavuria<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>HTML:</b><br><a href="#">Photosynthesis And Transpiration</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: iicba<br> <a href="#">Search ARES</a> for similar readings | The class DQB is projected or displayed at the front. Teacher reads aloud the driving question: 'How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die?' Students individually review the sticky notes/questions they added in Lessons 1–5. Each student selects ONE question from the DQB that they can now answer fully or partially using today's lesson. They write a 2–3 sentence answer on a new sticky note (different colour — e.g., green = 'I can now answer this') and place it over or beside the original question. Additionally, if today's lesson has raised a NEW question, students write it on a yellow sticky note and add it to the board. The class reviews 4–5 answered questions together, and the teacher draws a large arrow on the board timeline showing: Soil → Root → Stem → Leaf → ATMOSPHERE, marking 'Lesson 6 — LEAF' as the final node in the water journey. | Say: 'Look at our Driving Question Board. You have been adding questions since Lesson 1. Today, for the first time, we can answer the FULL journey — soil to atmosphere. Take 3 minutes. Find ONE question on the board that you can now answer. Write your answer on a GREEN sticky note and place it on the board.' After 3 minutes: 'Does anyone have a NEW question that today raised? Write it on yellow and add it.' Cold-call 3 students to read their green answers aloud. After each, ask the class: 'Do you agree? Is this a complete answer? What could make it stronger?' [WAIT 10 seconds.] Then point to the board timeline: 'Let us draw the complete journey together.' Build the arrow: Soil → Root hair (osmosis, L2) → Root cortex → Xylem vessel (L3–4) → Stem xylem (L5) → Leaf xylem → Mesophyll cells → Stomata → Atmosphere (L6). Say: 'This journey — and where it fails — is the answer to our driving question. Next lesson, we will add phloem and food transport to complete the picture.' | Progressive Disclosure on the DQB: The DQB functions as a cumulative public record of the class's evolving understanding. By physically moving from 'question' sticky notes to 'answer' sticky notes, students experience epistemic progress — they can SEE what they have figured out. The board timeline arrow makes the causal chain spatial and memorable. This strategy, named 'Storyline Mapping,' helps students see that science knowledge builds lesson by lesson toward a coherent explanation — not as isolated facts. | Teacher reads all new green sticky notes after class. Observable indicator: answer notes correctly reference stomata as the exit point and/or transpiration rate as the mechanism linking soil water to wilting. Red flag: green notes that say only 'water goes through stomata' without any reference to rate, conditions, or the Kibwezi context — these students need follow-up prompting at the start of Lesson 7. |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br>Source: Seavuria<br> <a href="#">Search ARES</a>                                                                                                                                                                                                                                                                                                                                                                                                               | Students turn to their cumulative transport model — a hand-drawn diagram they have been revising since Lesson 1, showing the full plant cross-section from root hair to leaf tip. Working individually, they add the following new structures and annotations to their model:<br>1. Leaf cross-section box (zoomed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Say: 'Open your transport model from Lesson 1. You are going to make it more complete than it has ever been. You have 10 minutes to add the leaf cross-section inset, the transpiration arrow, the four factors, and the Kibwezi annotation. I am looking for FOUR labelled structures in your leaf inset and the Kibwezi annotation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Cumulative Model Revision with Peer Review: The cumulative model strategy — used across all 8 lessons — externalises student thinking and makes conceptual growth visible over time. Each revision is additive: students do not start fresh but build on prior work, mirroring how scientific models evolve. Peer review                                                                                                                                                                                                          | Collect 5–6 models at random at the end of class. Observable proficiency indicators: (1) Leaf inset contains at least 4 correctly labelled structures including stomata and xylem. (2) Transpiration arrow exits through stomata — not through the cuticle or upper epidermis. (3) Kibwezi annotation correctly identifies the                                                                                          |

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| <p><a href="#">for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Photosynthesis And Transpiration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: iicba<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>inset): label upper epidermis, waxy cuticle, palisade mesophyll, spongy mesophyll with air spaces, lower epidermis, stomata with guard cells, xylem and phloem in vascular bundle.</p> <p>2. Arrow labelled 'water vapour exits as transpiration' pointing from stomata outward.</p> <p>3. Annotations for factors: next to the leaf, write 'Rate increases with: HIGH temperature, HIGH wind speed, LOW humidity, HIGH light intensity.'</p> <p>4. A red annotation connecting back to Kibwezi: 'At 2:00 pm in Kibwezi: all four factors are at maximum → transpiration rate &gt;&gt; water uptake rate → wilting.'</p> <p>Students who finish early add a 'comparison inset' showing the sukuma wiki under the mango tree — annotating how shade reduces light intensity and temperature, decreasing transpiration rate, explaining why it barely wilts.</p> <p>Peer review: each student swaps models with a partner. The partner checks for the 5 required structures and writes one 'strength' and one 'suggestion' on a sticky note attached to the model.</p> | <p>specifically.' Project a partially completed model on the board — show the empty leaf inset with dotted borders — say: 'Your leaf inset should look something like this frame. Fill it in using your own diagram from today's practical.'</p> <p>Circulate during model building. Stop at 5 minutes and say: 'Pause. Check — does your model show WHERE water exits the plant? If not, add that arrow now.' At 8 minutes: 'One more minute — make sure your Kibwezi annotation is written.' After 10 minutes: 'Swap with your partner. Read their model. Write one STRENGTH and one SUGGESTION on a sticky note. You have 3 minutes.' After peer review, cold-call 2 students: 'What was the best thing about your partner's model?' and 'What did you suggest they improve?' End with: 'Your model now shows the complete water journey. In Lesson 7, we will add the phloem — the food highway — and complete the full transport system.'</p> | <p>introduces the scientific practice of critique and revision (NGSS SEP 6: Constructing Explanations and Designing Solutions; SEP 7: Engaging in Argument from Evidence). Named for students: 'Scientists constantly revise their models as new evidence arrives — you are doing exactly that today.'</p> | <p>rate mismatch (transpiration exceeds uptake) as the cause of wilting. (4) At least 2 of the 4 transpiration factors are correctly listed. Criterion for mastery: all 4 indicators present. Criterion for reteaching flag: transpiration arrow exits from wrong location, OR Kibwezi annotation attributes wilting to 'no water in soil' (prior misconception not yet resolved).</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PHENOMENON CONNECTION: Could every student articulate — in their own words — why the maize in Kibwezi wilts in the afternoon even after morning watering? Did the CER task reveal students who still hold the misconception that 'watering = no wilting'? What will you do in Lesson 7's opening to address this?
2. PRACTICAL QUALITY: Did the stomatal impression technique work with the leaf species available? If sukuma wiki leaves gave unclear impressions, which alternative leaves in your school compound gave better results? Note this for next year. Did the nail varnish dry sufficiently in the 2–3 minute window, or do you need to adjust timing?

3. **DYE EVIDENCE:** Were students able to clearly identify stained xylem tissue from Lesson 5? If results were ambiguous, consider whether the dye concentration was sufficient (2 teaspoons of food colouring per 200 ml water is recommended) or whether stems were left in dye long enough (minimum 4 hours). Adjust Lesson 5 setup instructions accordingly.
4. **DIFFERENTIATION:** Did higher-ability students complete the sukuma wiki comparison inset (shaded plot)? Were lower-ability students supported in completing at least the 4 required leaf inset labels? Consider preparing a partially pre-labelled scaffold diagram for the next cohort.
5. **TIME MANAGEMENT:** The Observe phase (3 simultaneous observations) is the most time-pressured segment. Did 25 minutes feel sufficient? If not, consider moving the dye result recording to a warm-up activity before the lesson formally begins.
6. **DRIVING QUESTION PROGRESS:** Review the DQB green sticky notes. Are students' answers becoming more mechanistic (citing stomata, transpiration rate, xylem) or are they still descriptive ('water comes out of the leaf')? This tells you how ready students are for Lesson 7's phloem and pressure-flow content.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <b>What did I observe?</b>                   | Students observed water droplets inside a polythene bag tied around a leafy branch (sweating leaf bag demonstration), confirming that leaves release water vapour. They recorded results from the Lesson 5 coloured dye experiment, observing stained xylem tissue in stem cross-sections. They prepared stomatal impressions from sukuma wiki leaves using clear nail varnish and observed guard cells and stomatal pores under a hand lens, finding more stomata on the lower epidermis. They made cross-sections of the leaf midrib and identified the vascular bundle, epidermis layers, mesophyll, and waxy cuticle.                                                                                                                                                                                                                                                             |
| <b>What did I learn?</b>                     | Students learned that transpiration is the loss of water vapour from a plant, mainly through stomata in the lower epidermis of leaves. They learned that xylem vessels carry water from the roots all the way to the leaf mesophyll cells, where it evaporates and exits through stomata as water vapour. They identified four factors that increase the rate of transpiration: high temperature, high wind speed, low humidity, and high light intensity — all of which are present in Kibwezi on a dry afternoon. They explained why the farmer's maize wilts: transpiration rate in the hot, windy, dry afternoon exceeds the rate at which roots can absorb water from the soil, causing a water deficit inside the plant.                                                                                                                                                        |
| <b>How does this explain the phenomenon?</b> | Students can now explain the complete water journey from soil to atmosphere: water enters root hairs by osmosis, travels through root cortex cells, enters xylem vessels, is pulled up the stem by the transpiration pull (tension created as water evaporates from leaves), reaches leaf mesophyll cells through vein xylem, evaporates into air spaces, and exits through stomata as water vapour. The wilting of the Kibwezi maize is explained by a rate mismatch: on hot, dry, windy afternoons, stomata are open for photosynthesis, and the four transpiration factors combine to drive water loss far faster than roots can replace it, causing cells to lose turgor and the plant to wilt. The sukuma wiki under the mango tree barely wilts because shade reduces both light intensity and temperature — two of the four factors — dramatically slowing transpiration rate. |

## LESSON 7 (80 minutes): The Engine: Transpiration Pull, Cohesion-Tension, and Root Pressure

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Purpose</b>              | To investigate and explain the mechanisms that drive upward water movement in plants — transpiration pull (cohesion-tension) and root pressure — and to use quantitative data to partially resolve why the Kibwezi maize wilts even after watering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | Students will be able to: (a) describe the cohesion-tension mechanism as the primary driver of water movement up the xylem; (b) explain how transpiration pull is generated at the leaf surface; (c) describe root pressure as a supplementary driver of upward water movement; (d) define water potential and explain how gradients drive movement from soil → root → stem → leaf → atmosphere.                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Skills</b>               | Students will be able to: (a) set up and use an improvised potometer to measure water uptake rates under varied conditions; (b) collect, record, and graph quantitative data from the potometer investigation; (c) analyse graphs to identify patterns and draw evidence-based conclusions; (d) construct a written scientific explanation linking transpiration rate, water uptake rate, and wilting.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Attitudes</b>            | Students will: (a) appreciate precision and repeatability in data collection; (b) show curiosity about natural phenomena affecting Kenyan farmers; (c) demonstrate responsibility in handling shared laboratory equipment; (d) value the contribution of quantitative evidence in supporting or refuting a scientific explanation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Purpose in Storyline</b> | This is Lesson 7 of 8 in the evidence-gathering sequence. Students now have prior knowledge of xylem structure (Lesson 4), osmosis and root hair absorption (Lesson 5), and stomata function (Lesson 6). This lesson provides the mechanistic engine that connects all previous evidence: transpiration pull, cohesion-tension, and root pressure explain HOW water travels the full journey from soil to leaf. Students generate their own quantitative data using an improvised potometer, then use it to directly answer the anchoring phenomenon — the rate of transpiration in Kibwezi's hot, dry, windy afternoon exceeds the rate of water absorption, causing wilting even in watered soil. This lesson sets up Lesson 8, in which students will build their final cumulative model of plant transport. |
| <b>Safety Notes</b>         | Caution students to handle cut stems carefully — fresh cuts should be made under water using a razor blade or sharp knife; teacher should supervise all cutting. Warn students not to touch the lamp (heat source) directly. Petroleum jelly should not be placed near eyes or mouth. Students should wash hands after handling plant material. Ensure rubber tubing connections are secure before blowing or creating suction.                                                                                                                                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Students use an improvised potometer — assembled from a freshly cut leafy stem (e.g., a sprig of bougainvillea or a succulent herb available locally), rubber tubing, a calibrated capillary tube, and a ruler — to measure the rate of water uptake under four conditions: (1) control (still air, no lamp), (2) fan/blowing wind, (3) lamp directed at leaves, (4) petroleum jelly applied to leaf surfaces. Working in groups of four, they record the distance the air bubble moves in the capillary tube over two-minute intervals, calculate uptake rates (mm/min), and plot bar graphs. They then use the CLAIM–EVIDENCE–REASONING (CER) framework to construct an explanation of the cohesion-tension mechanism. Finally, they return to the Driving Question Board and post a sticky note answering: "Why does the maize wilt even after the farmer waters the soil?" This lesson partially resolves the anchoring phenomenon and prepares students for the final cumulative model in Lesson 8.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                       | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Seavuria<br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES for readings</a><br>Search terms:<br>plant transport<br>cohesion tension | Students are shown a diagram of a 3-metre-tall maize plant with soil at the base and a leaf at the top. The teacher poses: 'The farmer in Kibwezi watered her soil this morning. By 2 pm, the maize is wilting. The water is still in the soil — you can see it. What force is pulling water from the soil all the way up to the top leaf, 3 metres in the air? And what might be making that force stronger at 2 pm than at 6 am?' Students write individual predictions in their science notebooks using the sentence starter: 'I think water moves upward because... and I think the force is stronger at 2 pm because...' After 3 minutes of individual writing, students share predictions with a shoulder partner (Think-Pair-Share). The teacher cold-calls 4 pairs (not volunteers) to share, recording key prediction words on the board in two columns: FORCES PUSHING (e.g., 'roots push it up', 'pressure') and FORCES PULLING (e.g., 'leaves pull it', 'sun pulls water', 'evaporation'). Teacher does NOT correct predictions — all ideas are honoured and left on the board to be revisited. | 1. Display the tall maize diagram on the board or projector. Say verbatim: 'Before we do any investigation today, I want to know what YOU already think. No wrong answers — this is your honest prediction.' 2. Distribute science notebooks. Give students exactly 3 minutes of silent individual writing — use a visible timer. Do not allow discussion during this phase. 3. Say: 'Now share your prediction with your shoulder partner. You have 90 seconds each — Partner A speaks first, then Partner B.' 4. After Think-Pair-Share, cold-call exactly 4 pairs (use the class list, not raised hands). Ask each pair: 'Tell us ONE force you think is involved and why.' Record responses in the two columns (PUSHING / PULLING) on the board. 5. After all 4 pairs share, say: 'Notice that some of you think something PUSHES water up, and others think something PULLS it. Keep that question alive — our investigation today will give us data to test both ideas.' 6. Point to the DQB and say: 'We already have questions on our board about why maize wilts and how water gets to the top. Today we will gather quantitative — number-based — evidence to help us answer those questions.' | THINK-PAIR-SHARE with PREDICTION SORTING: Students activate prior knowledge from Lessons 4–6 (xylem structure, osmosis, stomata) and commit to a prediction before seeing evidence. Sorting predictions into PUSH vs. PULL creates a conceptual tension that motivates the investigation. Recording predictions publicly means students can compare their thinking to evidence at the end of the lesson, making conceptual change visible. | Teacher scans written predictions while circulating during individual writing time. Observable indicators of prior learning: (a) students reference osmosis or root pressure as a 'push' force (from Lesson 5); (b) students reference stomata or evaporation as a 'pull' (from Lesson 6); (c) students who cannot connect to prior lessons write vague predictions — teacher notes these students for targeted support during the Observe phase. Teacher does NOT grade predictions; they serve as a diagnostic baseline. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br><i>Related topic: not</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Groups of 4 receive a pre-assembled improvised potometer: a freshly cut leafy stem (bougainvillea sprig or locally available herb) inserted into rubber tubing connected to a calibrated capillary tube (marked in mm with a permanent marker and ruler), with the open end of the capillary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1. Before distributing equipment, demonstrate the potometer setup to the whole class (3 minutes). Say verbatim: 'Watch how the air bubble moves — that bubble is your measurement tool. The faster it moves toward the stem, the faster the stem is taking up water. This is your evidence.' 2. Distribute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | STRUCTURED DATA COLLECTION + CLASS DATA POOLING: Each group generates data for one or two conditions, but every student analyses the full class dataset. This mirrors authentic scientific practice (distributed investigation, shared data) and ensures all students                                                                                                                                                                      | Teacher checks (a) that results tables are correctly formatted with units (mm and mm/min); (b) that students calculate rate correctly (distance ÷ time, not just distance); (c) that bar graphs have labelled axes, a title, and correct scale. Students who annotate their graph with a sentence linking 'high                                                                                                                                                                                                            |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: Seavuria<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i></p> <p><a href="#">Search ARES for readings</a><br/> Search terms:<br/> plant transport<br/> cohesion tension</p> | <p>tube submerged in a small beaker of water. An air bubble is introduced into the capillary tube by briefly lifting the tube out of the water. Students observe the air bubble move as the stem takes up water. Each group tests TWO of the four conditions (conditions are distributed across groups so the whole class generates a full dataset): Condition A — Control (still air, no additional heat); Condition B — Fan or student blowing gently on the leaves for 2 minutes (simulates Kibwezi afternoon wind); Condition C — Desk lamp directed at leaves from 15 cm (simulates intense afternoon sun); Condition D — Petroleum jelly applied to the upper surface of all leaves (blocks stomata). For each condition, students record the distance the bubble moves (mm) every 2 minutes for 6 minutes, calculate the average rate (mm/min), and record in a results table. Groups then share data with the class on the board. ALL students copy the full class dataset and draw a bar graph (x-axis: conditions A–D; y-axis: water uptake rate in mm/min). Students annotate their graphs with one sentence explaining the pattern they observe.</p> | <p>equipment to groups. Say: 'You have 20 minutes. Your first job is to get a stable bubble and record your control reading. Do NOT apply the lamp or fan until you have your control data.' 3. Circulate every 3–4 minutes. At each group, crouch to eye level and ask one specific question: (a) 'What is your bubble doing right now — is that what you predicted?' (b) 'How are you making sure your reading is fair? What are you keeping the same?' (c) 'If the bubble is moving faster, what does that MEAN for water uptake?' 4. At the 18-minute mark, say: 'Two minutes to finish your last reading and calculate your average rate. Rate = total distance divided by total time.' 5. Invite one representative from each condition group to write their average rate on the class board. Give WAIT TIME of 10 seconds after all four values are posted, then ask: 'Before I say anything — what pattern do you notice in these four numbers? Write one sentence in your notebook.' Cold-call 3 students (not volunteers). 6. Say: 'Now draw your bar graph using the class data. X-axis: the four conditions. Y-axis: water uptake rate in mm/min. Title: Water Uptake Rate Under Different Conditions — Kibwezi Investigation.' Allow 8 minutes for graph drawing. 7. Ask: 'Write one sentence on your graph: which condition produced the HIGHEST water uptake rate, and what does that tell us about what drives water loss in Kibwezi at 2 pm?' Give 10 seconds WAIT TIME, then cold-call 2 students.</p> | <p>have quantitative evidence to use in the Explain phase. The bar graph transforms raw numbers into a visual pattern, making the relationship between environmental conditions (wind, heat, blocked stomata) and water uptake rate immediately visible and discussable.</p> | <p>uptake rate' to 'high transpiration' show readiness for the Explain phase. Students who write only descriptive annotations ('the bar is tallest for wind') need a bridging question: 'Why do you think the stem is taking up MORE water when the fan is blowing?'</p> |
| <p><b>Explain Phase</b></p>                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Using their potometer data,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1. Say: 'Put your graphs in front of</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>CLAIM–EVIDENCE–REASONING</p>                                                                                                                                                                                                                                              | <p>Teacher collects CER written</p>                                                                                                                                                                                                                                      |

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| <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES for readings</a><br/> Search terms:<br/> plant transport<br/> cohesion tension</p> | <p>students apply the CLAIM–EVIDENCE–REASONING (CER) framework to construct a written explanation. The teacher first provides a short guided explanation (10 minutes) of the cohesion-tension mechanism using a drawn diagram: (1) Water evaporates from the mesophyll cells through stomata (transpiration); (2) This lowers water potential in the leaf, pulling water from adjacent xylem cells; (3) Because water molecules COHERE (stick together via hydrogen bonds), the entire water column in the xylem is pulled upward like a chain; (4) This creates TENSION in the xylem — a negative pressure; (5) At the root end, water enters by osmosis from the soil, replenishing the column. Teacher also briefly explains ROOT PRESSURE: at night or in moist conditions, active uptake of mineral ions into root cells lowers the water potential of root cells, drawing water in by osmosis, which can push water upward (evidenced by guttation). Students then write their CER explanation independently: CLAIM — 'Water moves up the xylem because...'; EVIDENCE — at least two specific data points from their potometer investigation (with units); REASONING — 'This supports the cohesion-tension mechanism because...', explicitly linking the data to the mechanism. Students share CER explanations with a partner, give one piece of feedback using the stem: 'Your reasoning would be stronger if you explained WHY [condition] increased the rate...'</p> | <p>you — we are going to BUILD an explanation together using your data and some new vocabulary.' Draw the cohesion-tension diagram live on the board (do not use a pre-drawn poster — drawing it step-by-step slows the explanation to student thinking speed). 2. At each step of the diagram, PAUSE and ask a specific question: (a) After drawing stomata: 'From Lesson 6, what opens the stomata at 2 pm on a hot Kibwezi afternoon? Give me the word.' WAIT 10 seconds. Cold-call 1 student. (b) After drawing the water column: 'If water is evaporating from the top of this chain, what happens to the chain? Think about pulling a string from one end.' WAIT 12 seconds. Cold-call 2 students. (c) After drawing root entry: 'What process from Lesson 5 gets water INTO the root hair cell from the soil? Say the word.' WAIT 10 seconds. Cold-call 1 student. 3. Introduce the words COHESION and TENSION with a physical analogy: 'Imagine the water molecules in the xylem are like the students in this class holding hands in a line. If the student at the front is pulled forward, the whole line moves — that is cohesion-tension. The force doing the pulling is transpiration at the leaf.' 4. Say: 'Now write your CER. You have 10 minutes. Your CLAIM must start: Water moves up the xylem because... Your EVIDENCE must include at least TWO numbers from your potometer data. Your REASONING must use the words: cohesion, tension, transpiration, and water potential.' 5. After CER writing, say: 'Read your partner's REASONING section only. Use</p> | <p>(CER) FRAMEWORK: CER is a disciplinary literacy strategy that structures scientific argumentation. By requiring students to cite specific numerical evidence from THEIR OWN potometer data, the framework ensures the explanation is grounded in the lesson's investigation rather than being copied from a textbook. The peer feedback step makes reasoning quality visible and improvable. The return to initial predictions at the end makes conceptual change explicit and personally meaningful.</p> | <p>explanations at the end of class (or photographs them). Success criteria: (a) CLAIM correctly identifies transpiration pull (cohesion-tension) as the primary mechanism; (b) EVIDENCE cites at least two specific data points with units (e.g., 'Under wind conditions, uptake rate was 4.2 mm/min vs. 1.1 mm/min in the control'); (c) REASONING uses all four target vocabulary words (cohesion, tension, transpiration, water potential) correctly and logically links evidence to the mechanism. Students who include root pressure in their explanation and contrast it with cohesion-tension are demonstrating extended understanding. Students whose REASONING is circular ('it goes up because transpiration makes it go up') need the bridging question: 'What property of water molecules makes the pull of transpiration work all the way down to the roots?'</p> |
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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES</a> for readings<br/> Search terms: plant transport cohesion tension</p> | <p>Students return to the class Driving Question Board, which has been accumulating questions and evidence cards since Lesson 1. Each student is given one sticky note and asked to respond to the specific question: 'Why does the Kibwezi maize wilt even when the farmer has already watered the soil?' Students write their answer in 2–3 sentences, explicitly using the evidence from today's potometer investigation and the cohesion-tension mechanism. Students post their sticky notes on the DQB under the heading 'LESSON 7 ANSWER: Rate of Transpiration vs. Rate of Absorption.' The teacher then facilitates a 5-minute whole-class gallery walk where students read 4–5 sticky notes posted by peers. After the gallery walk, the teacher narrates the current state of the class's collective understanding by pointing to different cards on the DQB, showing how Lessons 4 → 5 → 6 → 7 have progressively built the answer to the driving question. Teacher explicitly names which parts of the driving question</p> | <p>1. Distribute one sticky note per student. Say verbatim: 'You have been investigating this question since Lesson 1 — why does the Kibwezi maize wilt even after watering? Today you have real data. Write your best answer in 2–3 sentences on this sticky note. You must include: (a) the word TRANSPIRATION, (b) the word ABSORPTION, and (c) at least one number from your potometer data.' Give 4 minutes of individual writing time. 2. Say: 'Post your note on the DQB under Lesson 7. Then do a silent gallery walk — read at least 4 notes that are NOT yours. In your notebook, write down one answer that is BETTER or MORE COMPLETE than yours, and explain why.' Give 5 minutes for the gallery walk. 3. Return to the front. Point to the DQB and narrate: 'In Lesson 4, we learned that xylem vessels are the pipelines. In Lesson 5, we learned that osmosis pulls water into the root hair. In Lesson 6, we learned that stomata open and close, controlling water loss. TODAY, we learned that transpiration pull —</p> | <p>DQB PROGRESSIVE NARRATIVE + GALLERY WALK: The DQB is a public record of the class's growing understanding. By physically pointing to how each lesson has added a piece to the answer, the teacher makes the storyline of the unit visible and coherent. The gallery walk activates social comparison — students evaluate their own explanations against peers' and identify what a stronger explanation looks like, which is both a metacognitive and argumentative practice.</p> | <p>Teacher reads all sticky notes after class. Observable success indicators: (a) student correctly identifies that transpiration RATE exceeds absorption RATE as the cause of wilting; (b) student references the cohesion-tension mechanism by name; (c) student includes at least one quantitative reference (even approximate). Students who write only 'the plant loses water through leaves' without referencing rates or mechanism need targeted follow-up at the start of Lesson 8. Teacher sorts sticky notes into three piles: Strong (all three criteria met), Developing (two criteria), Needs Support (one or fewer criteria) — these piles inform Lesson 8 grouping decisions.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | REMAIN unanswered (e.g., 'Why do some plants recover by evening but others die permanently?' — pointing forward to Lesson 8).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | the cohesion-tension mechanism — is the ENGINE that drives the whole journey.' (Trace the journey on a simple diagram as you speak.) 4. Say: 'But look — we still have unanswered questions on this board.' Point to cards asking: 'Why do SOME plants recover by evening?' and 'What happens to the xylem when the plant permanently wilts?' Say: 'Those questions belong to Lesson 8. Come ready.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Seavuria<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES for readings</a><br/> Search terms:<br/> plant transport cohesion tension</p> | <p>Students retrieve their cumulative transport model — a diagram of the full plant (roots, stem, leaves) that they have been adding to since Lesson 1. In this lesson, they make the following additions using a different coloured pen (to show what was added in Lesson 7): (1) Add ARROWS inside the xylem labelled 'cohesion-tension pull' pointing upward from root to leaf; (2) Add a label at the leaf showing 'transpiration — water exits via stomata → creates tension → pulls water column up'; (3) Add a small upward arrow at the root labelled 'root pressure (supplementary)'; (4) Add a new annotation box connected to the stomata: 'At 2 pm in Kibwezi: stomata open → high transpiration rate → tension in xylem increases → if soil water is insufficient, column breaks → wilting.' Students then write a one-sentence MODEL CLAIM at the bottom of their diagram: 'My model now explains wilting because...' Students share their updated model with a partner and use the sentence stem: 'One thing my model still does NOT explain is...' — this unsatisfied question is written on a new sticky note and added to the DQB as a 'bridge to</p> | <p>1. Say: 'Take out your transport model. You have been building this since Lesson 1. Today we add the ENGINE — the force that makes everything else work.' Give students 2 minutes to look over their existing model before adding anything. 2. Say: 'Use a NEW colour. Add three things: (1) cohesion-tension arrows in the xylem, (2) a transpiration label at the leaf connecting to the tension, (3) a root pressure label at the root. You have 8 minutes.' Circulate and check that arrows are directionally correct (upward) and that labels use the correct vocabulary. 3. After 8 minutes, say: 'Now write your MODEL CLAIM at the bottom: My model now explains wilting because... You have 2 minutes.' 4. Pair-share: Say: 'Show your partner your model. Tell them one thing your model NOW explains that it did NOT explain before Lesson 7. Then tell them one thing your model STILL does NOT explain.' Give 3 minutes. 5. Circulate and listen for unanswered questions. Common responses: 'I don't know why some plants recover by evening', 'I don't know what happens to the xylem when it</p> | <p>CUMULATIVE MODEL REVISION WITH COLOUR-CODED ANNOTATIONS: Using a new colour for each lesson's additions makes conceptual growth visible — students can literally see how their model has evolved across 7 lessons. The 'one thing my model still does NOT explain' prompt is a metacognitive strategy that honours incomplete understanding as a legitimate scientific state and creates forward motivation for Lesson 8. The MODEL CLAIM sentence requires students to synthesise the lesson's key idea into a single explanatory statement, which is a high-order cognitive task.</p> | <p>Teacher circulates during model revision and checks: (a) cohesion-tension arrows are pointing upward (not downward); (b) transpiration label is connected to the stomata at the leaf (not at the root); (c) root pressure label is at the root and described as 'supplementary' or 'a push force.' Students who draw arrows in the wrong direction or place transpiration at the root reveal a persistent misconception from prior lessons — teacher provides immediate individual correction using the chain-of-students analogy: 'The pull starts at the TOP of the chain — at the leaf — not at the bottom.' MODEL CLAIM sentences are collected (photographed or submitted) to assess whether students can synthesise the cohesion-tension mechanism into a coherent causal explanation.</p> |

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|  | Lesson 8' question. | <p>permanently wilts', 'I don't know how food moves — only water.'</p> <p>Say: 'Write those questions on a sticky note and add them to the DQB — those are exactly the questions Lesson 8 will address.'</p> <p>6. Close by saying: 'Your model is almost complete. In Lesson 8, you will add the final layer — translocation of food through the phloem — and you will present your full model as your final explanation of the Kibwezi phenomenon.'</p> |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) DATA QUALITY: Did all groups obtain a clear, moving bubble in their potometer? If not, what set-up issues arose (leaky connections, no visible bubble, stem blocked) and how will you address these for a repeat or for Lesson 8 review? (2) CER QUALITY: When you review students' written CER explanations, how many correctly linked their numerical data to the cohesion-tension mechanism (not just described the data)? Which students defaulted to textbook language without referencing their own evidence — and what does this tell you about their relationship to inquiry? (3) MISCONCEPTIONS OBSERVED: Did any students persist in believing that roots 'pump' water up (root pressure as the PRIMARY mechanism rather than supplementary)? How did you address this — and do you need to revisit it at the start of Lesson 8? (4) DQB STICKY NOTES: After sorting sticky notes into Strong / Developing / Needs Support, what proportion of the class can correctly articulate that wilting results from transpiration rate exceeding absorption rate? How will you use this information to group students in Lesson 8? (5) PHENOMENON CONNECTION: Did students consistently connect their potometer data back to Kibwezi — or did the investigation feel disconnected from the farmer's problem? If the latter, consider opening Lesson 8 by re-reading the phenomenon aloud and asking students to point to the specific part of their model that now explains each observation (afternoon wilting, recovery by evening, shade protecting the sukuma wiki).

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <b>What did I observe?</b>                   | Students observed an air bubble moving through a calibrated capillary tube in an improvised potometer, measuring how fast a cut leafy stem took up water under four conditions: control (still air), wind (fan/blowing), heat (lamp), and blocked stomata (petroleum jelly). They observed that wind and heat conditions produced the fastest bubble movement (highest water uptake rates), while blocking the stomata dramatically slowed the bubble. Students also observed class-pooled data posted on the board and drew bar graphs comparing uptake rates across all four conditions.                       |
| <b>What did I learn?</b>                     | Students learned that water moves up the xylem primarily through the cohesion-tension mechanism: transpiration at the leaf surface lowers water potential, creating tension that pulls the cohesive water column upward from roots to leaves. Root pressure provides a supplementary upward push. The rate of water uptake increases when transpiration is accelerated by wind and heat — conditions that match a hot, dry Kibwezi afternoon. Blocking stomata (petroleum jelly) sharply reduces water uptake, confirming that transpiration through stomata is the primary driver of the entire upward journey. |
| <b>How does this explain the phenomenon?</b> | Students explained, using the CER framework and their potometer data, that the Kibwezi maize wilts even after watering because the rate of transpiration in the hot, windy afternoon exceeds the rate at which water can be absorbed from the soil and transported up the xylem. When transpiration rate is very high, the tension in the xylem increases; if water in the soil is insufficient or absorption                                                                                                                                                                                                    |

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|  | is too slow, the cohesive water column cannot be replenished fast enough, causing cells to lose turgor and the plant to wilt. Students updated their cumulative transport model with cohesion-tension arrows, a transpiration label at the leaf, and a root pressure label at the root. |
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## LESSON 8 (80 minutes): The Return Journey: Translocation and Final Model Build

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To consolidate understanding of plant transport by examining phloem translocation through the pressure-flow hypothesis, completing a final annotated whole-plant transport model, and constructing a full evidence-based scientific explanation of the Kibwezi wilting phenomenon.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) explain the structure and function of phloem in translocating sugars and amino acids from source to sink; (b) describe the pressure-flow hypothesis as the mechanism driving translocation; (c) distinguish between xylem transport (water and mineral salts, root to leaf, transpiration pull) and phloem transport (sucrose and amino acids, source to sink, pressure flow); (d) explain the evidence from girdling experiments that supports phloem function.                                                                                                                                                                                                                                                                                                                                        |
| <b>Skills</b>               | The learner should be able to: (a) construct and annotate a complete, cumulative whole-plant transport model integrating osmosis, xylem, phloem, stomatal transpiration, and the pressure-flow hypothesis; (b) analyse photographic evidence of girdled trees to infer the role of phloem; (c) write a structured, evidence-based scientific explanation of the anchoring phenomenon; (d) evaluate and compare their Lesson 1 initial model against their final model to identify conceptual growth.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Attitudes</b>            | The learner should: (a) appreciate that scientific understanding is built incrementally through evidence gathered across multiple investigations; (b) recognise the real-world relevance of plant transport biology to food security and farming in semi-arid regions like Kibwezi, Makueni County; (c) value collaborative sensemaking and the role of peer critique in refining scientific models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Key Inquiry Question</b> | How are water and mineral salts transported in plants? How are manufactured food substances transported in plants?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Purpose in Storyline</b> | This is the final lesson in the Plant Transport unit. Students have previously built understanding of osmosis at root hairs (L1–L2), xylem structure and cohesion-tension theory (L3–L4), transpiration and stomatal regulation (L5–L6), and the role of mineral deficiency in plant failure (L7). In this lesson, students complete the 'return journey' — examining how sugars manufactured in leaves travel back down through phloem to roots, fruits, and growing tips. This closes the loop on the Kibwezi phenomenon: students can now explain not only why the maize wilts (water loss exceeds uptake via transpiration pull and xylem) but also why some plants die (phloem failure, sugar starvation of roots, collapse of the whole transport system). The final cumulative model and written explanation represent mastery of all unit learning outcomes. |
| <b>Safety Notes</b>         | No hazardous chemicals are used in this lesson. If girdling demonstrations use real plant stems, ensure students do not use sharp tools unsupervised. Photographs of girdled mango or eucalyptus trees from Kenyan farms may be used in lieu of live specimens. Ensure all printed materials are returned and recycled responsibly (PCIs: Environmental Conservation).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### B. LESSON OVERVIEW

In this final lesson of Sub-Strand 2.2, students examine photographic evidence of girdled trees — where the bark (phloem) is removed in a ring — and observe that sugars accumulate above the girdle while roots below starve and die. This provides direct empirical evidence for downward phloem transport. Students then apply the pressure-flow (mass flow) hypothesis to explain how a high-pressure source (photosynthesising leaf cells loaded with sucrose) pushes sugars through sieve tubes towards low-pressure sinks (roots, fruits, growing tips). Students use this understanding to complete their final cumulative whole-plant transport model, annotating xylem and phloem pathways, osmosis at root hairs, stomatal transpiration, cohesion-tension, and pressure flow. Groups then present their models and co-construct a full class scientific explanation of the Kibwezi wilting phenomenon — answering every question on the Driving Question Board. The lesson closes with a model comparison (Lesson 1 vs. Lesson 8), a teacher-facilitated DQB review confirming all questions are answered, and individual written Final Explanations submitted as a unit

summative task.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                    | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Seavuria<br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES for readings</a><br>Search terms:<br>plant transport<br>phloem xylem | Students are shown two photographs side-by-side on the board (or printed handouts): (1) a healthy mango tree on a farm near Mombasa, and (2) a mango tree whose bark has been removed in a complete ring around the trunk (girdling) — showing a swollen, bulging ring of tissue above the cut, wilting and dead roots below, and abundant fruit set just above the girdle. Students work individually for 3 minutes to write predictions in their science notebooks: 'What do you think is happening in this tree? Where is water moving? Where is food moving? Will this tree survive?' They then share predictions with a partner (Think-Pair-Share, 2 minutes). The teacher uses cold-calling (4 students) to surface predictions on the board under two columns: 'Water story' and 'Food story.' Students also re-read the Kibwezi scenario card (distributed in Lesson 1) and write one sentence: 'What question about the Kibwezi maize can I now almost answer — and what do I still not know?' | Display the two mango tree photographs. Say: 'Look carefully at both images. Do not call out yet — I want everyone to think privately first.' [WAIT TIME: 15 seconds]. Say: 'Write your prediction in your notebook. You have 3 minutes. Use these sentence starters: I think water is... I think food is... I think this tree will...' [3 minutes individual writing]. After Think-Pair-Share, cold-call 4 students — two who look confident and two who appear uncertain: 'Amina, tell us your water story.' / 'Omondi, do you agree with Amina or do you have a different idea?' / 'Wanjiru, what does your partner think will happen to the roots?' / 'Kipchoge, what question do you still have?' Record all predictions verbatim on the board under 'Water story' and 'Food story' — do NOT correct errors yet. Say: 'We are going to use today's evidence to test every single one of these predictions. Keep your notebook open — you will be revising these predictions before the lesson ends.' | Think-Pair-Share with Cold-Call: This strategy activates prior knowledge from Lessons 1–7 (osmosis, xylem, transpiration) and creates cognitive dissonance — students who understand xylem will predict water moves upward, but the girdling image challenges them to think about a separate downward food-transport pathway. Recording predictions publicly creates accountability and a reference point for revision. | Observable indicator: Students write at least two distinct predictions — one about water movement and one about food/sugar movement — showing they are beginning to distinguish between two separate transport systems. Teacher scans notebooks during cold-call and notes any students who conflate xylem and phloem (e.g., 'water carries the food down too') — these students will be targeted for support during Phase 3. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br><i>Related topic: not</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Activity A — Girdling Evidence Analysis (15 minutes): Students work in groups of 4 to analyse a structured evidence card containing: (1) annotated photographs of a girdled eucalyptus tree from a plantation near Kericho — showing swelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Distribute evidence cards to groups. Say: 'You have 15 minutes. Your group must reach consensus on all four questions — every member must be able to explain the group's answer. I will cold-call one person from each group, and it may not be the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Evidence-Based Reasoning + Physical Analogy Model: The girdling evidence card moves students from prediction to empirical reasoning — they must use concentration data and anatomical diagrams to infer function. The pressure-flow                                                                                                                                                                                     | Observable indicator: Group written answers show correct identification of phloem as the sucrose-transporting tissue, correct directionality (source leaf → sink root/fruit), and use of the terms 'concentration gradient' and 'pressure.' Teacher listens during                                                                                                                                                            |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: Seavuria<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i></p> <p><a href="#">Search ARES for readings</a><br/> Search terms:<br/> plant transport<br/> phloem xylem</p> | <p>above girdle, dead roots below; (2) a data table showing sucrose concentration above vs. below girdle in a girdled bean plant after 48 hours (above girdle: 4.2%, below girdle: 0.3%); (3) a diagram of phloem sieve tube cells and companion cells. Groups answer guided questions: 'Where did the sucrose accumulate? Why can it not cross the girdle? What does this tell you about which tissue carries sugar? In which direction does sugar normally travel?' Activity B — Pressure-Flow Model Demonstration (10 minutes): The teacher demonstrates the pressure-flow hypothesis using a simple physical model: two beakers connected by a glass tube (representing sieve tubes). Beaker A (source, leaf) contains concentrated sucrose solution dyed red; Beaker B (sink, root) contains dilute sucrose solution. A semi-permeable membrane separates each beaker from a water reservoir. Students observe water entering Beaker A by osmosis (pressure builds), pushing red solution toward Beaker B. Students sketch the model in their notebooks and label: source, sink, high pressure, low pressure, direction of flow, osmosis.</p> | <p>person who wrote the answer.' [Circulate, spending 2–3 minutes per group. Listen for: 'the food goes down' / 'the bark carries sugar' / 'the tree starves at the bottom'.] After 15 minutes, cold-call one reporter per group: 'Group 3 — Nyambura, in one sentence, what does the girdling data tell us about which tissue carries sugar?' [WAIT TIME: 10 seconds before accepting response.] After all groups report, say: 'Let me show you a model of HOW the sugar is pushed.' Conduct the pressure-flow demonstration. Narrate aloud: 'This red solution is sucrose leaving a photosynthesising leaf cell — this is our source. Watch what happens when osmosis builds pressure here.' After the demonstration, say: 'Sketch this in your notebook right now. Label source, sink, and draw an arrow showing direction of flow. You have 90 seconds.' [WAIT TIME: 90 seconds, then cold-call 2 students to share their sketches via document camera or verbal description.]</p> | <p>physical demonstration makes an abstract molecular mechanism visible and tangible, directly connecting the chemistry of sucrose concentration gradients to the mechanics of mass flow. Both activities are anchored to the Kibwezi context: if phloem fails (due to drought stress or pest damage to bark), sugars cannot reach roots — the root system weakens, water uptake collapses, and the plant dies rather than recovers.</p> | <p>circulation for misconceptions: 'sugar travels in xylem alongside water' (target for Phase 3 direct instruction). Notebook sketches of pressure-flow model should show correctly labelled arrows; teacher checks 4–5 notebooks during the 90-second sketch time.</p>                                                                                                                |
| <p><b>Explain Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem, Transpiration [3D Animation]</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Seavuria<br/> <a href="#">Search ARES for similar videos</a></p>                                                      | <p>Part A — Direct Instruction Consolidation (10 minutes): The teacher delivers a focused, structured explanation of the pressure-flow (mass flow) hypothesis, explicitly connecting each step to the Kibwezi maize plant: (1) Sucrose is produced by photosynthesis in mesophyll cells of the maize leaf (source). (2) Sucrose is actively loaded into phloem sieve tubes — water</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>For Part A, use the board to draw a maize plant outline — annotate live as you narrate the pressure-flow steps. Say: 'Watch my pen. I am drawing what is happening right now inside a maize plant in Kibwezi.' After each of the 4 steps, pause and ask: 'Can someone predict what happens next?' [WAIT TIME: 10 seconds, cold-call 1 student.] After Part A, say: 'Now I want you to be the expert. Open</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Structured Note-Making + Phenomenon Reconnection: The T-table forces students to explicitly articulate the distinction between xylem and phloem across multiple dimensions, preventing the persistent misconception that both systems do the same job. Reconnecting all three questions to Kibwezi at the end of Phase 3 ensures that declarative knowledge (pressure-flow</p>                                                        | <p>Observable indicator: T-table entries correctly distinguish xylem (water + minerals, dead cells, cohesion-tension, upward) from phloem (sucrose + amino acids, living sieve tubes + companion cells, pressure flow, source-to-sink). Kibwezi written answers should reference specific mechanisms: transpiration exceeding uptake (Q1), stomatal closure and overnight recovery</p> |

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| <p> <b>READING:</b><br/>No closely matching reading in the ARES library for this activity.</p> <p> <a href="#">Search ARES for readings</a><br/>Search terms:<br/>plant transport<br/>phloem xylem</p> | <p>follows by osmosis, raising hydrostatic pressure. (3) At the sink (root cells, cob, growing tip), sucrose is unloaded and used; water leaves by osmosis, lowering pressure. (4) The pressure gradient drives mass flow from source to sink. Part B — Compare and Contrast (10 minutes): Students complete a structured T-table in their notebooks comparing XYLEM and PHLOEM across six criteria: tissue type, contents transported, direction of transport, driving force, living/non-living cells, evidence used to establish function. Students work individually for 5 minutes, then peer-check with a partner. Part C — Connecting to Kibwezi (5 minutes): Students re-read the Kibwezi scenario. The teacher poses three focus questions written on the board: (1) 'Why does the maize wilt even though soil has water?' (2) 'Why do some plants die while others recover by evening?' (3) 'Why does the sukuma wiki under the mango tree barely wilt?' Students write 2-sentence answers to each question individually, drawing explicitly on evidence from Lessons 1–8.</p> | <p>your notebook to a blank page. Draw this T-table. You have 5 minutes — silence please, individual work.' [5 minutes.] After peer-check, cold-call 3 students to share one row each of the T-table: 'Chebet, read us your row on driving force.' / 'Mwangi, do you agree? Is there anything to add or correct?' [WAIT TIME: 10 seconds.] Correct any remaining misconception that xylem and phloem carry the same substances in the same direction. For Part C, say: 'These three questions are the heart of our entire unit. Write your best answers now — use specific evidence from our experiments and readings. No copying from a neighbour.' [5 minutes individual writing.] Cold-call 3 students (one per question): 'Akinyi, give us your answer to question 2 — why do some plants die while others recover?' [WAIT TIME: 12 seconds.]</p> | <p>mechanism) is immediately linked to explanatory power (why the farmer's maize behaves as it does). This is consistent with NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 2 (Developing and Using Models).</p>                                                                                                                                                                               | <p>(Q2), reduced transpiration demand in shade (Q3). Teacher collects 6 random notebooks at end of Phase 3 for rapid review before Phase 4.</p>                                                                                                                                                                                                                                                                                         |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/><a href="#">12 Plant Transport Xylem And Phloem</a>, <a href="#">Transpiration [3D Animation]</a><br/><i>Related topic: not an exact match for this lesson.</i></p>                                            | <p>The class gathers around the Driving Question Board (physical or digital) that has been built incrementally across Lessons 1–7. The board displays all student-generated questions from the unit, organised under three clusters: (A) Water and mineral uptake, (B) Water movement through the plant, (C) Food transport and plant survival. The teacher reads each question aloud. For each question, the class identifies which lesson(s)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Stand at the DQB. Say: 'This board has been our scientific record for eight lessons. Let us honour the work by going through every question.' Read each question clearly and slowly. For each: 'Who remembers which lesson helped us answer this? What was the evidence?' [WAIT TIME: 10 seconds, cold-call different students for each question — aim for at least 12 different students across the full</p>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Driving Question Board Synthesis: The DQB is a visible, cumulative record of the class's epistemic journey. Physically moving questions from 'wondering' to 'explained' externalises metacognition — students see their own intellectual growth as a concrete, spatial record. Requiring students to cite specific evidence for each answer reinforces the norm that scientific claims must be supported by data, not just</p> | <p>Observable indicator: Students can correctly identify the specific lesson, activity, and mechanism that answers each DQB question without prompting. Personal reflection entries show genuine metacognitive awareness — students identify a specific concept (not just 'everything') that was difficult, and articulate specifically what changed their thinking. Teacher listens for unprompted cross-lesson connections (e.g.,</p> |

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| <p>Source: Seavuria<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> No closely matching reading in the ARES library for this activity.</p> <p> <a href="#">Search ARES</a><br/> for readings<br/> Search terms:<br/> plant transport<br/> phloem xylem</p>                                                | <p>provided the answer, which evidence activity (experiment, girdling data, celery staining, etc.) was decisive, and which mechanism explains it. Students physically move question cards from the 'Still wondering' column to the 'We can explain this now' column — or add a written answer sticky note. Any questions that remain unanswered are acknowledged honestly and recorded as 'Questions for further investigation' (connecting to the nature of science). The teacher highlights the journey: 'In Lesson 1 we could not explain why the Kibwezi maize wilted. Look at how many questions we can now answer.' Students spend 3 minutes writing a personal DQB reflection in their notebooks: 'The question I found hardest to answer was... because... Now I understand it because...'</p> | <p>DQB review.] When a question is moved to 'answered,' say: 'What experiment or observation gave us the right to say that?' — insist students cite specific evidence, not just recall vocabulary. If a question is partially answered, say: 'Let us be honest scientists — we can partially explain this. What do we still not know?' Record unanswered questions in a different colour. After the full DQB review, give 3 minutes of silent reflection writing. Cold-call 2 students to share their reflections: 'Ouma, which question was hardest for you? Tell us why.' / 'Zawadi, what changed in your thinking the most?' [WAIT TIME: 10 seconds each.] Say: 'This is exactly how real scientists work — they build understanding question by question, experiment by experiment. Prof. Wangari Maathai did not understand why deforestation caused soil erosion in one day. It took years of careful observation. You have done something similar in 8 lessons.'</p> | <p>assertions. The personal reflection writing deepens individual ownership of the learning.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>'the celery staining in Lesson 6 showed us the xylem path, and today the girdling showed us the phloem path is separate') as evidence of deep integration.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">12 Plant Transport Xylem And Phloem. Transpiration [3D Animation]</a><br/> Related topic: not an exact match for this lesson.<br/> Source: Seavuria<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>READING:</b><br/> No closely matching reading</p> | <p>Part A — Final Model Construction (15 minutes): Each group receives a large blank outline of a maize plant (roots, stem, leaf, cob) on A3 paper. Groups have access to their Lesson 1 initial model for comparison. Using coloured pens (blue for water/xylem, red for sugars/phloem, green for osmosis arrows, orange for transpiration), groups construct their final annotated model. Required annotations: (1) Osmosis at root hair cells — water potential gradient, direction of movement; (2) Xylem vessel pathway — contents (water + mineral salts), driving force (transpiration pull /</p>                                                                                                                                                                                               | <p>Distribute A3 maize outlines and coloured pens. Return Lesson 1 models to each group. Say: 'You have 15 minutes. Your final model must tell the complete story — from a raindrop soaking into the soil in Kibwezi, to that water evaporating from a stomata on the top leaf, AND the journey of a sucrose molecule from a leaf mesophyll cell down to a root tip. Every arrow must be labelled. Every colour must be used correctly.' [Circulate, spending 2–3 minutes per group. Prompt with: 'Where does the pressure-flow begin in this plant?' / 'Show me exactly where the transport fails</p>                                                                                                                                                                                                                                                                                                                                                                      | <p>Cumulative Model Revision + Evidence-Based Scientific Explanation Writing: The final model build is the capstone sensemaking activity of the entire unit. Requiring colour-coded annotation of both xylem and phloem on the same plant diagram forces students to integrate two distinct transport systems into a unified whole — they cannot treat them separately. The Kibwezi wilting annotation specifically requires students to apply their model to the original phenomenon, demonstrating transfer. The written Final Explanation (NGSS SEP 6: Constructing Explanations)</p> | <p>Observable indicators — Final model: All five required annotations are present and correctly labelled; colour coding is consistent; Kibwezi wilting annotation correctly identifies the breakdown point (transpiration rate exceeds xylem supply rate; and/or phloem failure starves roots; and/or stomata fail to close due to loss of guard cell turgor). Written Final Explanation: Minimum 150 words; all 10 required terms used correctly in context; Kibwezi environmental factor named and mechanistically linked to transport failure. Model comparison sentence: Identifies a specific</p> |

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| <p>in the ARES library for this activity.</p> <p><a href="#">Search ARES for readings</a></p> <p>Search terms:<br/>plant transport<br/>phloem xylem</p> | <p>cohesion-tension), direction (root → leaf); (3) Stomatal transpiration — guard cell mechanism, environmental triggers (heat, wind, drought); (4) Phloem sieve tube pathway — contents (sucrose + amino acids), driving force (pressure-flow / mass flow), direction (source leaf → sink root/cob); (5) Kibwezi wilting annotation — mark exactly where the transport system breaks down in the wilting scenario and why. Part B — Group Presentations (10 minutes): Each group presents their final model (2 minutes per group, 4 groups). One designated 'challenger' from another group asks one question. Part C — Individual Written Final Explanation (10 minutes): Working individually and in silence, each student writes a structured scientific explanation (minimum 150 words) answering the full driving question: 'How does water and food travel through an entire plant — from soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die?' The explanation must reference: osmosis, water potential, xylem, cohesion-tension, transpiration, stomata, phloem, pressure-flow hypothesis, source, sink, and at least one specific environmental factor from Kibwezi (heat, drought, semi-arid conditions). Part D — Model Comparison (3 minutes): Students place their Lesson 1 model beside their Lesson 8 model. They write one sentence: 'My biggest scientific growth in this unit was...' These are collected as evidence of learning progression.</p> | <p>when the maize wilts.' / 'Is that arrow pointing the right direction for phloem?'] During presentations, instruct challengers: 'Your question must start with: How does your model explain...' After presentations, say: 'Now close your group work. Individual silence. This final explanation is yours alone.' [10 minutes writing.] At the 5-minute mark, say quietly: 'Make sure you have named at least one specific environmental condition in Kibwezi that disrupts transport.' After writing, say: 'Place your Lesson 1 model beside your Lesson 8 model. Look at both. Write one sentence beginning: My biggest scientific growth in this unit was...' Collect all final explanations and model comparison sentences. Close the lesson by reading the driving question aloud one final time and saying: 'Every word of that question — you can now answer. That is biology.'</p> | <p>requires students to synthesise mechanism, evidence, and context into a coherent scientific argument — this is the highest cognitive demand of the unit. The model comparison (L1 vs. L8) makes learning progression visible and supports metacognitive reflection, a key competency in Kenya's CBC framework.</p> | <p>conceptual shift (not generic), demonstrating genuine metacognitive awareness. Teacher uses final explanations as summative assessment evidence for the unit.</p> |
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## D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions:

1. **CONCEPTUAL MASTERY:** Did students correctly distinguish between xylem and phloem transport in their final models — including directionality, driving forces, and contents? Were there persistent confusions (e.g., phloem moving upward only, or water travelling in phloem) that indicate a need to revisit in Grade 11?
2. **PHENOMENON CLOSURE:** Were students able to articulate a complete, mechanistically accurate explanation of the Kibwezi wilting scenario in their Final Explanations? Could they explain all three elements: (a) why wilting occurs despite watered soil, (b) why some plants recover by evening while others die, and (c) why the shaded sukuma wiki is protected? If students struggled with (b) or (c), consider adding a dedicated lesson on stomatal regulation and shade effects in future iterations.
3. **MODEL QUALITY:** Comparing Lesson 1 models to Lesson 8 models — was the growth visible and significant? Were students able to articulate their own growth in the model comparison sentence? If Lesson 1 models were very sparse, consider adding a 'checkpoint model' at Lesson 4 to make progression more gradual and visible.
4. **DQB COMPLETENESS:** Were all student-generated questions on the DQB answered by the end of this lesson? If questions remained unanswered, note them here and consider whether they should be addressed in Sub-Strand 2.3 (Gaseous Exchange and Respiration) or flagged as genuine open scientific questions.
5. **DIFFERENTIATION:** Were all learners able to engage with the pressure-flow hypothesis, or did some students remain at a surface-level 'sugars go down' understanding? Consider whether a visual animation or a further guided reading on phloem loading (active transport of sucrose into sieve tubes) would support struggling learners.
6. **TIME MANAGEMENT:** This lesson is content-rich. If time was short, which phase was compressed? The written Final Explanation (Phase 5C) should never be sacrificed — if time pressure is acute, shorten group presentations (Phase 5B) to 1 minute per group.
7. **KENYA CONTEXT:** Did students connect the biology to real farming decisions in semi-arid regions like Kibwezi or the wider Makueni County? If not, consider inviting a guest speaker (local agricultural extension officer) or assigning a follow-up task: 'Write a letter to the Kibwezi farmer explaining what is happening to her maize and what she could do differently, using biology.'

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <b>What did I observe?</b> | Students observed annotated photographs and concentration data from girdling experiments (eucalyptus trees near Kericho; bean plant sucrose data) showing accumulation of sugars above the girdle and starvation below, providing direct evidence that phloem carries sucrose downward from source to sink. Students also observed a physical pressure-flow demonstration using two beakers, concentrated and dilute sucrose solutions, semi-permeable membranes, and water — watching mass flow driven by osmotic pressure gradients.                                                                                                                                                                                                                                              |
| <b>What did I learn?</b>   | Students learned that phloem sieve tubes, assisted by companion cells, transport sucrose and amino acids from photosynthetic source cells (leaves) to metabolic sink cells (roots, fruits, growing tips) by mass flow driven by a hydrostatic pressure gradient — the pressure-flow (mass flow) hypothesis. Students consolidated the full plant transport system: osmosis drives water into root hairs down a water potential gradient → cohesion-tension pulls water up xylem → transpiration at stomata drives the pull → sucrose is loaded into phloem at source → pressure-flow moves sucrose to sinks → osmosis unloads water at sink. Students applied this integrated understanding to construct a complete annotated model and to explain why the Kibwezi maize wilts, why |

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|                                              | some plants die, and why shaded sukuma wiki is protected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>How does this explain the phenomenon?</b> | Students explained the anchoring phenomenon — 'The Wilting Maize of Kibwezi' — in full: (1) Despite watered soil, the rate of transpiration through stomata on a hot, dry Kibwezi afternoon exceeds the rate of water uptake and xylem transport, creating a water deficit in leaf cells, causing loss of turgor and wilting. (2) Plants near the dry seasonal riverbed wilt sooner because soil water potential is lower and root uptake is less efficient, meaning the xylem system cannot replace transpired water fast enough; if guard cells lose turgor and fail to close stomata, or if the phloem system is compromised by drought stress, the plant cannot recover and dies. (3) Sukuma wiki under the mango tree experiences lower transpiration demand because shade reduces leaf temperature and the vapour pressure gradient between leaf and air — stomata need not open as wide, less water is lost, the xylem supply is sufficient, turgor is maintained, and the plant does not wilt. |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
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| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |